

Black-Box Separation Between Pseudorandom Unitaries, Pseudorandom Isometries, and Pseudorandom Function-Like States

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Abstract

Pseudorandom functions (PRFs) are one of the most fundamental primitives in classical cryptography. On the other hand, in quantum cryptography, it is possible that PRFs do not exist but their quantum analogues could exist, and still enabling many applications including SKE, MACs, commitments, multiparty computations, and more. Pseudorandom unitaries (PRUs) [Ji, Liu, Song, Crypto 2018], pseudorandom isometries (PRIs) [Ananth, Gulati, Kaleoglu, Lin, Eurocrypt 2024], and pseudorandom function-like state generators (PRFSGs) [Ananth, Qian, Yuen, Crypto 2022] are major quantum analogs of PRFs. PRUs imply PRIs, and PRIs imply PRFSGs, but the converse implications remain unknown. An important open question is whether these natural quantum analogues of PRFs are equivalent. In this paper, we partially resolve this question by ruling out black-box constructions of them:

1. There are no black-box constructions of $O(\log \lambda)$ -ancilla PRUs from PRFSGs.
2. There are no black-box constructions of $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch from PRFSGs.
3. There are no black-box constructions of $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch from PRIs with $\Omega(\lambda)$ stretch.

Here, $O(\log \lambda)$ -ancilla means that the generation algorithm uses at most $O(\log \lambda)$ ancilla qubits. PRIs with $s(\lambda)$ stretch is PRIs mapping λ qubits to $\lambda + s(\lambda)$ qubits. To rule out the above black-box constructions, we construct a unitary oracle that separates them. For the separations, we construct an adversary based on the quantum singular value transformation, which would be independent of interest and should be useful for other oracle separations in quantum cryptography.

*A part of the work was done when the last author visited UCSB.

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1 Introduction

Pseudorandom functions (PRFs) [GGM86] are among the most fundamental primitives in classical cryptography. PRFs formalize the hardness of distinguishing certain functions from truly random functions, and have numerous important applications including IND-CPA secret-key encryption (SKE) [GGM86] and EUF-CMA message authentication codes (MAC) [GGM84]. Moreover, PRFs are existentially equivalent to one-way functions (OWFs) [GGM86, HILL99, GKL93, Lev85], which indicates that PRFs are existentially equivalent to all Minicrypt primitives and are implied by almost all computationally-secure cryptographic primitives.

In quantum cryptography, on the other hand, it is possible that PRFs do not exist but quantum analogs of PRFs could exist [JLS18, AQY22, BBSS23, AGKL24, LQS⁺24, BM24, Kre21, KQT24], and many applications are still possible from them [JLS18, AQY22, MY22]. Pseudorandom unitaries (PRUs) [JLS18], pseudorandom isometries (PRIs) [AGKL24], and pseudorandom function-like state generators (PRFSGs) [AQY22] are major quantum analogs of PRFs. A PRU is a family $\{U_k\}_k$ of unitaries implementable in quantum polynomial-time (QPT) that are computationally indistinguishable from Haar random unitaries. A PRI is a family $\{\mathcal{I}_k\}_k$ of QPT implementable isometries that are computationally indistinguishable from Haar random isometries.¹ A PRFSG is a QPT algorithm that, on input a classical key k and a classical bit string x , outputs a quantum state $|\phi_k(x)\rangle$ that is computationally indistinguishable from Haar random states. PRUs, PRIs, and PRFSGs could exist even if PRFs do not exist [Kre21, KQT24]. Moreover, PRFSGs imply various primitives and applications [JLS18, MY22, Yan22, AQY22, MYY24, KT24].

PRUs imply PRIs, and PRIs imply PRFSGs [AGQY22]. However, although they are natural quantum analogs of PRFs, it remains an open question whether the reverse implication holds. This naturally raises the following question:

Are PRUs, PRIs, and PRFSGs equivalent?

Given their crucial roles in quantum cryptography, an important open problem is to determine whether these natural quantum analogues of PRFs are equivalent.

1.1 Our Results

In this paper, we partially resolve the above open problem by ruling out black-box constructions for restricted cases. The first result is the following:

Theorem 1.1. *There is no black-box construction of non-adaptive and $O(\log \lambda)$ -ancilla PRUs from PRFSGs.*

Here, a black-box construction is defined as follows [CM24, CCS24].

Definition 1.2 (Black-Box Construction of Non-Adaptive PRUs from PRFSGs). *We say that non-adaptive PRUs can be constructed from PRFSGs in a black-box way if there exist QPT algorithms $C^{(\cdot, \cdot)}$ and $R^{(\cdot, \cdot)}$ such that both of the following two conditions are satisfied:*

1. *Black-box construction: For any QPT algorithm G satisfying the correctness of PRFSGs² and for any its unitary implementation $\tilde{G}$,³ $C^{\tilde{G}, \tilde{G}^\dagger}$ satisfies the correctness of non-adaptive PRUs.⁴*

¹Here, Haar random isometry acts as $|\psi\rangle \mapsto U(|\psi\rangle |0\dots 0\rangle)$, where U is Haar random unitary.

²We say that a QPT algorithm satisfies the correctness of PRFSGs if it takes bit strings k and x as input and outputs a pure state.

³In general G is a CPTP map. The CPTP map G can be implemented by applying a unitary $\tilde{G}$ on a state and tracing out some qubits. A unitary implementation of G is such a unitary $\tilde{G}$.

⁴We say that a QPT algorithm satisfies the correctness of non-adaptive PRUs if it takes a classical bit string k and a quantum state as input and applies a unitary on the input state.

2. *Black-box security reduction:* For any QPT algorithm G satisfying the correctness of PRFSGs, any its unitary implementation $\tilde{G}$, any adversary $\mathcal{A}$ that breaks the security of $C^{\tilde{G}, \tilde{G}^\dagger}$, and any unitary implementation $\tilde{\mathcal{A}}$ of $\mathcal{A}$, it holds that $R^{\tilde{\mathcal{A}}, \tilde{\mathcal{A}}^\dagger}$ breaks the security of G .

In this definition, unitary implementations and their inverses are queried⁵. There are other variants of black-box constructions. For example, only unitary implementations are queried, and their inverses are not queried. Alternatively, instead of unitary implementations, isometry implementations are queried. Definition 1.2 contains these variants [CM24, CCS24], and therefore it captures general black-box constructions.

Non-adaptive PRUs are a weaker variant of PRUs where the adversary can query the oracle only non-adaptively. $O(\log \lambda)$ -ancilla PRUs are PRUs that can be implemented in QPT using at most $O(\log \lambda)$ ancilla qubits. Because $(O(\log \lambda)$ -ancilla) PRUs imply non-adaptive (and $O(\log \lambda)$ -ancilla) PRUs, we have the following as a corollary:

Corollary 1.3. *There is no black-box construction of $O(\log \lambda)$ -ancilla PRUs from PRFSGs.*

We also point out that PRFSGs in Theorem 1.1 are quantumly-accessible and adaptively-secure ones, which is the strongest version of PRFSGs in the following sense: Recall that a PRFSG is a QPT algorithm G that, on input a classical key k and a classical bit string x , outputs a quantum state $|\phi_k(x)\rangle$ that is computationally indistinguishable from Haar random states. More precisely, the computational indistinguishability means that for any QPT adversary $\mathcal{A}$,

$$\left| \Pr_{k \leftarrow \{0,1\}^\lambda} [1 \leftarrow \mathcal{A}^{G(k, \cdot)}(1^\lambda)] - \Pr_{\mathcal{H}} [1 \leftarrow \mathcal{A}^{\mathcal{H}}(1^\lambda)] \right| \leq \text{negl}(\lambda), \quad (1)$$

where $\mathcal{H}$ is the following oracle: sample a Haar random state $|\psi_x\rangle$ for each x independently in advance, and when x is queried, return $|\psi_x\rangle$. Quantumly-accessible means that $\mathcal{A}$ can query superpositions of x . Adaptively-secure means that $\mathcal{A}$ can query the oracle adaptively. We can define variants of PRFSGs where the queries are only non-adaptive or classical ones. Clearly, quantumly-accessible and adaptively-secure PRFSGs are stronger than them.

Next, we show separation for PRIs.

Theorem 1.4. *There are no black-box constructions of non-adaptive and $O(\log \lambda)$ -ancilla PRUs from PRIs with $\Omega(\lambda)$ stretch.*

Theorem 1.5. *There is no black-box construction of non-adaptive and $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch from PRIs with $\Omega(\lambda)$ stretch.*

Here, $\{\mathcal{I}_k\}_k$ is called PRI with $s(\lambda)$ stretch if it is a family of QPT implementable isometries from λ qubits to $\lambda + s(\lambda)$ qubits and it is computationally indistinguishable from Haar random isometries. The notion of black-box construction used in Theorems 1.4 and 1.5 is defined similarly to Definition 1.2. Non-adaptive PRIs are a weaker variant of PRIs in which the adversary is restricted to making only non-adaptive oracle queries. $O(\log \lambda)$ -ancilla PRIs with s stretch are PRIs with s stretch that can be implemented in QPT using at most $s(\lambda) + O(\log \lambda)$ ancilla qubits.⁶ Because $(O(\log \lambda)$ -ancilla) PRIs imply non-adaptive (and $O(\log \lambda)$ -ancilla) PRIs, we have the following as a corollary:

Corollary 1.6. *There is no black-box construction of ancilla-free PRUs from PRIs with $\Omega(\lambda)$ stretch. In addition, there is no black-box construction of $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch from PRIs with $\Omega(\lambda)$ stretch.*

⁵In this paper, we do not consider a query to controlled-operation, transpose, and complex conjugate.

⁶The use of s ancilla qubits is necessary for PRIs with s stretch because it maps λ qubits to $\lambda + s(\lambda)$ qubits.

Our main results are Theorems 1.1, 1.4 and 1.5, but they are derived from the following technical results:

Theorem 1.7 (Theorem 3.3, Informal). *There exists a unitary oracle $\mathcal{O}$ such that PRFSGs exist but non-adaptive and $O(\log \lambda)$ -ancilla PRUs do not exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$.*

Theorem 1.8 (Theorem 6.1, Informal). *There exists a unitary oracle $\mathcal{O}$ such that PRFSGs exist but non-adaptive and $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch do not exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$.*

Theorem 1.9 (Theorem 7.1, Informal). *There exists a unitary oracle $\mathcal{O}$ such that PRIs with $\Omega(\lambda)$ stretch exist but non-adaptive and $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch do not exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$.*

Here the existence and non-existence of primitives relative to oracles mean the following.

Definition 1.10. *Let $\mathcal{O}$ be a unitary oracle. We say that a primitive exists relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$ if there exists a QPT algorithm $C^{(\cdot, \cdot)}$ such that both of the following two conditions are satisfied:*

- $C^{\mathcal{O}, \mathcal{O}^\dagger}$ satisfies the correctness of the primitive.
- $C^{\mathcal{O}, \mathcal{O}^\dagger}$ satisfies the security of the primitive against any QPT adversary $\mathcal{A}^{\mathcal{O}, \mathcal{O}^\dagger}$ that can query $\mathcal{O}$ and $\mathcal{O}^\dagger$.

In Appendix A.1, we will explain that this oracle separation implies the impossibility of the black-box construction. A high-level overview of our proofs of Theorems 1.7 to 1.9 will be explained in Section 1.2.

In the above definition, Definition 1.10, the query to both $\mathcal{O}$ and $\mathcal{O}^\dagger$ are allowed. If the query to only $\mathcal{O}$ is allowed, and that to $\mathcal{O}^\dagger$ is not allowed, what we can rule out is not the black-box construction of Definition 1.2, but a more restricted one where C queries only $\tilde{G}$ and R queries only $\tilde{A}$. Because our Theorem 1.7 shows the oracle separation in terms of Definition 1.10, we can exclude the general black-box constructions, which is an important advantage of our results⁷.

Theorems 1.7 to 1.9 also indicate that all primitives that are known to be implied by PRFSGs or PRIs (such as PRSGs, private-key quantum money, OWSGs, OWPuzzs, EFI pairs, SKE, commitments, MAC, etc.) also exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$. However, some caution is needed for the existence of IND-CPA SKE with quantum ciphertexts and EUF-CMA MAC (with unclonable tags), because known constructions of these primitives from PRFSGs [AQY22] query the inverse of a unitary implementation of PRFSGs. One advantage of our result, Theorem 1.7, is that the security of PRFSGs holds against adversaries that query not only $\mathcal{O}$ but also $\mathcal{O}^\dagger$. Because of this advantage, the known constructions of IND-CPA SKE with quantum ciphertexts and EUF-CMA MAC (with unclonable tags) from PRFSGs automatically imply their existence relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$. (For details, see Appendix A.2.)

Several new ideas and techniques are used to show Theorem 1.7, many of which are of independent interest and should be useful for other applications in quantum cryptography. In particular, for the oracle separation, we construct a direct attack to PRUs. To the best of our knowledge, this is the first time that a direct attack to PRUs has been constructed. All previous results that break PRUs first reduced PRUs to PRSGs and then broke PRSGs [Kre21, AGQY22] by using the shadow tomography [Aar19, HKP20]. As we will explain later, our key idea for constructing the direct attack to PRUs is to leverage the quantum singular-value transformation (QSVT) [GSLW19]. To our knowledge, this is the first time that QSVT has been used to separate quantum cryptographic primitives. We believe that QSVT should be useful for other applications in quantum cryptography.

Finally, our results are summarized in Figure 1.

⁷However, our oracle separations do not rule out the case when C queries $\tilde{G}^\top$, $\tilde{G}$, or R queries $\mathcal{A}^\top$, $\tilde{A}$, where $(\cdot)^\top$ denotes the transpose, and $(\cdot)$ denotes the complex conjugate. This is because queries to $\mathcal{O}^\top$ or $\tilde{\mathcal{O}}$ are not allowed in Theorems 1.7 to 1.9. We expect that our separations can be extended to the case when the query to $\mathcal{O}^\top$ and $\tilde{\mathcal{O}}$ are allowed using a similar technique in [Zha25].

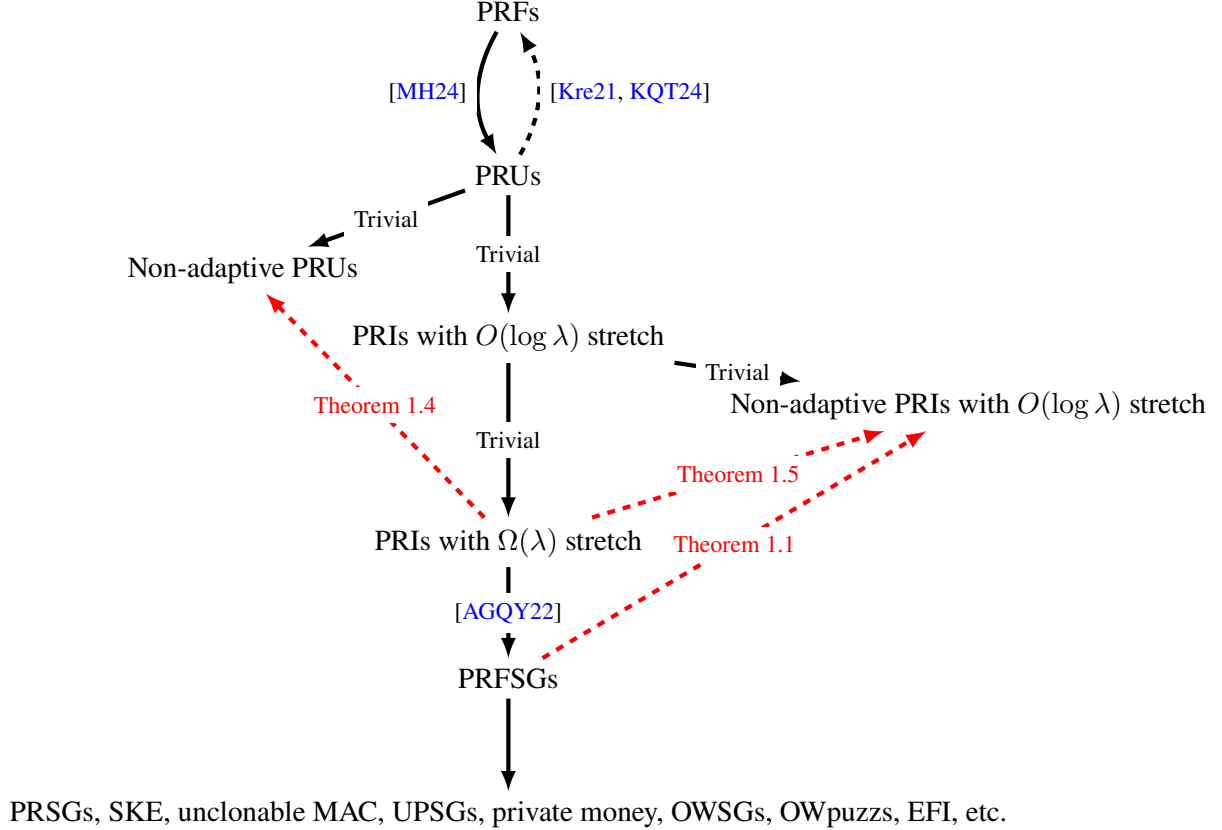


Figure 1: A summary of our results and known results. An arrow from primitive A to primitive B indicates that A implies B. A dashed arrow from A to B indicates that there is no black-box construction from A to B. A red dashed arrow from A to B indicates that there is no black-box construction from A to $O(\log \lambda)$ -ancilla B. SKE means IND-CPA SKE with quantum ciphertexts. Unclonable MAC means EUF-CMA MAC with unclonable tags. Private money means private-key quantum money schemes.

1.2 Technical Overview

As we have mentioned, our main results Theorems 1.1, 1.4 and 1.5 are obtained from the technical result Theorems 1.7 to 1.9. In this subsection, we will overview the high-level idea of the proof of Theorem 1.7. Theorems 1.8 and 1.9 can be shown in a similar idea.

Our goal is to construct a unitary oracle $\mathcal{O}$ such that PRFSGs exist but PRUs do not relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$. Our oracle $\mathcal{O}$ consists of two oracles $\mathcal{S}$ and $\mathcal{U}$. $\mathcal{S}$ is used to construct PRFSGs. $\mathcal{U}$ is an oracle that solves UnitaryPSPACE-complete problem [RY22, BEM⁺23], which is used to break PRUs.

Constructing PRFSGs. The oracle $\mathcal{S} := \{\mathcal{S}_{k,x}\}_{k,x}$ is a set of unitary oracles $\mathcal{S}_{k,x}$. Each $\mathcal{S}_{k,x}$ works as follows:

- Sample a Haar random state $|\psi_{k,x}\rangle$.
- If the input is $|0\rangle|0\dots 0\rangle$, return $|1\rangle|\psi_{k,x}\rangle$.

- If the input is $|1\rangle|\psi_{k,x}\rangle$, return $|0\rangle|0\dots 0\rangle$.
- For other inputs, do nothing.

In other words, the oracle $\mathcal{S}_{k,x}$ “swaps” $|0\rangle|0\dots 0\rangle$ and $|1\rangle|\psi_{k,x}\rangle$. Similar oracles were considered in [BMM⁺24, BCN24, CCS24]. With this $\mathcal{S}$, we construct a PRFSG $G^{\mathcal{S}}$ as follows:

1. On input (k, x) , query $|0\rangle|0\dots 0\rangle$ to $\mathcal{S}_{k,x}$ in order to get $|1\rangle|\psi_{k,x}\rangle$.
2. Output $|\psi_{k,x}\rangle$.

Intuitively, it is clear that $G^{\mathcal{S}}$ satisfies the security of PRFSGs, because its output is Haar random states. In fact, we can show the security of $G^{\mathcal{S}}$ based on the proof template of [Kre21]. However, the template cannot be directly used in this case, and some careful re-investigations are required. First, define

$$\text{Adv}(\mathcal{S}) := \Pr_k[1 \leftarrow \mathcal{A}^{\mathcal{S}_k, \mathcal{S}}] - \Pr_{\{|\vartheta_x\rangle\}_{x \leftarrow \sigma}}[1 \leftarrow \mathcal{A}^{\mathcal{T}, \mathcal{S}}], \quad (2)$$

where $\mathcal{S}_k := \{\mathcal{S}_{k,x}\}_x$, σ is a Haar random state measure and $\mathcal{T}$ is the following oracle.

1. For each x , sample a Haar random state $|\vartheta_x\rangle$ independently in advance.
2. When x is queried, it returns $|\vartheta_x\rangle$.

This $\text{Adv}(\mathcal{S})$ is the advantage of the adversary $\mathcal{A}$ in the security game of PRFSGs when $\mathcal{S}$ is chosen. (Actually, we have to consider not $\mathcal{A}^{\mathcal{S}}$ but $\mathcal{A}^{\mathcal{S}, \mathcal{S}^\dagger, \mathcal{U}, \mathcal{U}^\dagger}$. However, because $\mathcal{S}^\dagger = \mathcal{S}$ and $\mathcal{U}^\dagger$ can be simulated by querying $\mathcal{U}$, we have only to consider $\mathcal{A}^{\mathcal{S}, \mathcal{U}}$. Moreover, our argument below is about unbounded $\mathcal{A}$, and therefore we can also ignore $\mathcal{U}$.) Our goal is to show that $|\text{Adv}(\mathcal{S})|$ is small with high probability over $\mathcal{S}$. To that end, we first show

$$|\mathbb{E}_{\mathcal{S}} \text{Adv}(\mathcal{S})| \leq \text{negl}(\lambda). \quad (3)$$

This is shown by using the technique of [Kre21], which is based on the BBBV theorem [BBBV97]. However, the technique of [Kre21] cannot be directly used because in his case each $\mathcal{S}_k$ is a Haar random unitary, while in our case not. Fortunately, we can confirm that his proof also holds for our $\mathcal{S}_k$, and therefore we can show Equation (3) in a similar way. From Equation (3), we want to show our goal that $|\text{Adv}(\mathcal{S})|$ is small with high probability over $\mathcal{S}$ via the following concentration inequality:⁸

$$\Pr_{\mathcal{S}}[|\text{Adv}(\mathcal{S}) - \mathbb{E}_{\mathcal{S}'} \text{Adv}(\mathcal{S}')| \geq \delta] \leq e^{-O(\delta/L^2)}. \quad (4)$$

This concentration inequality holds if $\text{Adv}(\mathcal{S})$ is L -Lipshitz. By a straightforward calculation, we confirm it. In this way, we can show that our constructed PRFSG $G^{\mathcal{S}}$ is secure against $\mathcal{A}^{\mathcal{S}}$.

⁸ $\text{Adv}(\mathcal{S})$ is a function of unitary, because each Haar random state in $\mathcal{S}$ can be replaced with a Haar random unitary applied on $|0\dots 0\rangle$.

Breaking PRUs. Here we describe how to break ancilla-free PRUs, since the argument generalizes to $O(\log \lambda)$ -ancilla PRUs with postselection. Let $\mathcal{S}$ be the unitary oracle introduced above. Let $\mathcal{U}$ be the **UnitaryPSPACE**-complete oracle. Let $F^{\mathcal{S}, \mathcal{U}}$ be a QPT algorithm that, on input k and a state, applies a unitary U_k on the state. (Actually, we have to consider $F^{\mathcal{S}, \mathcal{S}^\dagger, \mathcal{U}, \mathcal{U}^\dagger}$, but as we have mentioned above, we can ignore $\mathcal{S}^\dagger$ and $\mathcal{U}^\dagger$.) Our goal is to show that $F^{\mathcal{S}, \mathcal{U}}$ cannot satisfy the security of PRUs. In other words, we construct a QPT adversary $\mathcal{A}^{\mathcal{S}, \mathcal{U}}$ that distinguishes the query to $F^{\mathcal{S}, \mathcal{U}}$ and that to the oracle that applies Haar random unitaries.

To show it, we first define two states

$$\rho_0 := \mathbb{E}_{U \leftarrow \mu} (U^{\otimes \ell} \otimes I) |\Phi\rangle \langle \Phi| (U^{\otimes \ell} \otimes I)^\dagger, \quad (5)$$

and

$$\rho_1 := \mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} (U_k^{\otimes \ell} \otimes I) |\Phi\rangle \langle \Phi| (U_k^{\otimes \ell} \otimes I)^\dagger. \quad (6)$$

Here, μ is the Haar measure over λ -qubit unitaries, U_k is a λ -qubit unitary, $\ell := \lceil \log |\mathcal{K}_\lambda| \rceil$, $\mathcal{K}_\lambda$ is the key space, $|\Phi\rangle := \frac{1}{\sqrt{2^{\ell\lambda}}} \sum_{x \in \{0,1\}^{\ell\lambda}} |x\rangle |x\rangle$ is the $\ell\lambda$ -qubit maximally entangled state. The adversary can generate ρ_0 if it queries the Haar random oracle, while it can generate ρ_1 if it queries $F^{\mathcal{S}, \mathcal{U}}$. Therefore, if ρ_0 and ρ_1 can be distinguished by using the **UnitaryPSPACE**-complete oracle $\mathcal{U}$, the adversary can break the PRU.

The question is therefore how to distinguish ρ_0 and ρ_1 by using $\mathcal{U}$? There is one issue here. Each U_k can depend on $\mathcal{S}$, because the PRU generator $F^{\mathcal{S}, \mathcal{U}}$ can query $\mathcal{S}$, while the **UnitaryPSPACE**-complete oracle $\mathcal{U}$ (that $\mathcal{A}$ queries) is independent of $\mathcal{S}$. When $\mathcal{S}$ acts on small number of qubits, $\mathcal{A}$ can get its classical information by querying $\mathcal{A}$'s $\mathcal{S}$ many times and doing the process tomography [HKOT23], and can send the classical information to $\mathcal{U}$ as is done in [Kre21]. However, when $\mathcal{S}$ acts on large number of qubits, this strategy does not work, because the process tomography is no longer efficient. Our key observation is that when $\mathcal{S}$ acts on large number of qubits, it almost does not cause any effect, because $\mathcal{S}$, which swaps only $|0\dots 0\rangle$ and Haar random states, is almost the identity operation.⁹

Hence, we introduce $\{U'_k\}_k$ that is the same as $\{U_k\}_k$ except that $\mathcal{S}$ acting on small number of qubits is simulated by the classical information obtained via the process tomography. Then, if we define another state

$$\rho_2 := \mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} ((U'_k)^{\otimes \ell} \otimes I) |\Phi\rangle \langle \Phi| ((U'_k)^{\otimes \ell} \otimes I)^\dagger, \quad (7)$$

our goal is to distinguish ρ_0 and ρ_2 by using $\mathcal{U}$ since ρ_1 is statistically close to ρ_2 . Let Q be the projection onto the support of ρ_2 . Clearly, $\text{Tr}[Q\rho_2] = 1$. On the other hand, as we will explain later, $\text{Tr}[Q\rho_0]$ is negligible. Therefore, if we can implement Q , we can distinguish ρ_0 and ρ_2 . The fact that $\text{Tr}[Q\rho_0]$ is negligible can be shown from the following lemma.

Lemma 1.11 (Lemma 5.8, Informal). *For the above Q and ρ_0 ,*

$$\text{Tr}[Q\rho_0] \leq \text{negl}(\lambda). \quad (8)$$

How can we implement Q with $\mathcal{U}$? Our novel idea is to use the singular-value discrimination (SVD) algorithm, which is a concrete example of quantum singular-value transformation (QSVT) [GSLW19]. Let M be any positive matrix such that there exists a unitary V that satisfies $M = (|0\dots 0\rangle \otimes I)V(|0\dots 0\rangle \otimes I)$.

⁹This step of our proof requires the ancilla-free condition. As mentioned earlier, we can relax this condition to the $O(\log \lambda)$ -ancilla condition by using postselection. Whether the same result can be established without this assumption remains an open question.

Such an encoding of a matrix to a unitary is called a block encoding [GSLW19]. The SVD algorithm, which can query V , can solve the following promise problem: Given a single-copy quantum state ξ , decide whether the support of ξ is in the support of M or the support of ξ is orthogonal to that of M . Because our goal is to implement Q , which decides whether a given state is in the support of ρ_2 , we have only to take $M = \rho_2$. It is known that a block encoding of any quantum state ξ can be efficiently implementable by using a unitary W satisfying that $W|0\dots 0\rangle$ is a purification of ξ [vAG19]. Thus if we take $M = \rho_2$, we can efficiently implement its block encoding. Then if we run the SVD algorithm on input ρ_0 or ρ_2 , we can distinguish ρ_0 and ρ_2 . The SVD algorithm can be realized in quantum polynomial space [GSLW19], and therefore Q can be realized by querying $\mathcal{U}$.

Recall that our original goal is to distinguish ρ_0 from ρ_1 . Since ρ_1 is statistically close to ρ_2 , the above algorithm can also distinguish ρ_0 from ρ_1 . In summary, therefore, a QPT adversary $\mathcal{A}^{\mathcal{S}, \mathcal{U}}$ can break the PRU $F^{\mathcal{S}, \mathcal{U}}$.

1.3 Related Works

Comparison with the concurrent work [BHMV25]. The concurrent work by [BHMV25] also separates non-adaptive and ancilla-free PRUs from PRFSGs. Their separation oracle, which they call unitary common Haar function-like state (CHFS) oracle, is the same as our separation oracle. However, they use a different technique to break non-adaptive and ancilla-free PRUs.

Comparison with the previous works. There are several works that separate Microcrypt primitives. In addition to the works we have already mentioned [Kre21, CCS24, BCN24, BMM⁺24], there are other three papers. [CM24] separated multi-query secure quantum digital signatures from PRUs. [AGL24] separated quantum computation classical communication (QCCC) primitives from PRFSGs. [GMMY24] constructed oracles such that QCCC primitives exist but $\mathbf{BQP} = \mathbf{QCMA}$, and quantum lightning [Zha19] exist but $\mathbf{BQP} = \mathbf{QMA}$ relative to the oracles. These separation results are incomparable with our work.

2 Preliminaries

2.1 Basic Notations

This paper uses the standard notations of quantum computing and cryptography. For bit strings x and y , (x, y) denotes their concatenation. We use λ as the security parameter. $[n]$ means the set $\{1, 2, \dots, n\}$. For any set S , $x \leftarrow S$ means that an element x is sampled uniformly at random from the set S . We write negl as a negligible function and poly as a polynomial. QPT stands for quantum polynomial time. For an algorithm $\mathcal{A}$, $y \leftarrow \mathcal{A}(x)$ means that the algorithm $\mathcal{A}$ outputs y on input x .

We use $I := |0\rangle\langle 0| + |1\rangle\langle 1|$ as the identity on a single qubit. For the notational simplicity, we sometimes write $I^{\otimes n}$ just as I when the dimension is clear from the context. For a vector $|\psi\rangle$, we define its norm as $\| |\psi\rangle \| := \sqrt{\langle \psi | \psi \rangle}$. For any matrix A , we define the p -norm $\|A\|_p := (\text{Tr}[(A^\dagger A)^{p/2}])^{1/p}$. In particular, we call it the trace norm when $p = 1$ and the Frobenius norm when $p = 2$. For any matrix A , the operator norm $\|\cdot\|_\infty$ is defined as $\|A\|_\infty := \max_{|\psi\rangle} \sqrt{\langle \psi | A^\dagger A | \psi \rangle}$, where the maximization is taken over all pure states $|\psi\rangle$. id denotes the identity channel, i.e., $\text{id}(\rho) = \rho$ for any state ρ . For two channels $\mathcal{E}$ and $\mathcal{F}$ that take d dimensional states as inputs, we say $\|\mathcal{E} - \mathcal{F}\|_\diamond := \max_{|\psi\rangle} \|(\text{id} \otimes \mathcal{E})(|\psi\rangle\langle\psi|) - (\text{id} \otimes \mathcal{F})(|\psi\rangle\langle\psi|)\|_1$ is the diamond norm between $\mathcal{E}$ and $\mathcal{F}$, where the maximization is taken over all d^2 dimensional pure states.

$\mathbb{L}(d)$ denotes the set of all $d \times d$ matrices. The set (or group) of d -dimensional unitary matrices and states are denoted by $\mathbb{U}(d)$ and $\mathbb{S}(d)$, respectively. μ_d and σ_d denote the Haar measure over $\mathbb{U}(d)$ and $\mathbb{S}(d)$, respectively. For $U \in \mathbb{U}(d)$, $\text{c-}U := |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$ is the controlled- U . $|\Omega_d\rangle := \frac{1}{\sqrt{d}} \sum_{x \in [d]} |x\rangle |x\rangle$ is the maximally entangled state. For $U \in \mathbb{U}(d)$, $U(\cdot)U^\dagger$ denotes the channel that maps $\rho \mapsto U\rho U^\dagger$ for all d -dimensional state ρ .

2.2 Useful Facts

Here we introduce several useful facts that we will use.

Lemma 2.1 (Gentle Measurement Lemma [Win99, Wat18]). *Let ρ be a quantum state, and $0 \leq \epsilon \leq 1$. Let M be a matrix such that $0 \leq M \leq I$ and*

$$\text{Tr}[M\rho] \geq 1 - \epsilon. \quad (9)$$

Then,

$$\left\| \rho - \frac{\sqrt{M}\rho\sqrt{M}}{\text{Tr}[M\rho]} \right\|_1 \leq \sqrt{\epsilon}. \quad (10)$$

The following is a well-known fact. For its proof, see [Bha13].

Lemma 2.2 (Hölder's Inequality [Bha13]). *Let A and B be matrices of the same size. Then, for any $p > 0$, we have $\|AB\|_p \leq \|A\|_p \|B\|_\infty$. Moreover, we have $\|AB\|_\infty \leq \|A\|_\infty \|B\|_\infty$*

We need the following lemma when we use the concentration inequality which we introduce later.

Lemma 2.3 (Lemma 28 in [Kre21]). *Let $\mathcal{A}^U$ be a quantum algorithm that makes T queries to $U \in \mathbb{U}(d)$ and its inverse. Then, $f(U) = \Pr[1 \leftarrow \mathcal{A}^U]$ is $2T$ -Lipschitz in the Frobenius norm, i.e., $|f(U) - f(V)| \leq 2T\|U - V\|_2$ for all $U, V \in \mathbb{U}(d)$.*

By applying the triangle inequality, we have the following.

Lemma 2.4. *Let $f, g : \mathbb{U}(d) \rightarrow \mathbb{R}$ be L -Lipschitz functions in the Frobenius norm, i.e., $|f(U) - f(V)| \leq L\|U - V\|_2$ and $|g(U) - g(V)| \leq L\|U - V\|_2$ for any $U, V \in \mathbb{U}(d)$. Then, $f + g$ is $2L$ -Lipschitz. Namely, for any $U, V \in \mathbb{U}(d)$,*

$$|f(U) + g(U) - f(V) - g(V)| \leq 2L\|U - V\|_2. \quad (11)$$

Regarding the Frobenius norm, we use the following.

Lemma 2.5. *Let $U, V \in \mathbb{U}(d)$. Then, for any $|\psi\rangle \in \mathbb{S}(d)$,*

$$\|U|\psi\rangle\langle\psi|U^\dagger - V|\psi\rangle\langle\psi|V^\dagger\|_2 \leq 2\|U - V\|_2 \quad (12)$$

Proof of Lemma 2.5. We obtain the inequality as follows:

$$\begin{aligned} & \|U|\psi\rangle\langle\psi|U^\dagger - V|\psi\rangle\langle\psi|V^\dagger\|_2 \\ & \leq \|U|\psi\rangle\langle\psi|U^\dagger - V|\psi\rangle\langle\psi|U^\dagger\|_2 + \|V|\psi\rangle\langle\psi|U^\dagger - V|\psi\rangle\langle\psi|V^\dagger\|_2 && \text{(By the triangle inequality)} \\ & = \|(U - V)|\psi\rangle\langle\psi|U^\dagger\|_2 + \|V|\psi\rangle\langle\psi|(U - V)^\dagger\|_2 \\ & = \|(U - V)|\psi\rangle\langle\psi|U^\dagger\|_2 + \|(U - V)|\psi\rangle\langle\psi|V^\dagger\|_2 && \text{(By } \|A^\dagger\|_2 = \|A\|_2\text{)} \\ & \leq \|U - V\|_2 \|\psi\rangle\langle\psi|U^\dagger\|_\infty + \|(U - V)\|_2 \|\psi\rangle\langle\psi|V^\dagger\|_\infty && \text{(By Hölder's inequality, Lemma 2.2)} \\ & \leq 2\|U - V\|_2, \end{aligned} \quad (13)$$

where in the last inequality we have used $\|\psi\rangle\langle\psi|W\|_\infty \leq \|\psi\rangle\langle\psi|\|_\infty \|W\|_\infty \leq 1$ for any $W \in \mathbb{U}(d)$ from Lemma 2.2. $\square$

The following lemma is implicitly shown in [Kre21].¹⁰

Lemma 2.6 (Lemma 31 in [Kre21]). *Let $\mathcal{D}$ be a distribution over $\mathbb{U}(d)$. Suppose that $\mathcal{A}$ is a quantum algorithm that queries $U = (U_1, \dots, U_N) \in \mathbb{U}(d)^N$ and $O \in \mathbb{U}(d)$. We see U as $\sum_{n \in [N]} |n\rangle\langle n| \otimes U_n$. For fixed $U \in \mathbb{U}(d)^N$, define*

$$\text{Adv}(\mathcal{A}, U) := \Pr_{k \leftarrow [N]} [1 \leftarrow \mathcal{A}^{U_k, U}] - \Pr_{O \leftarrow \mathcal{D}} [\mathcal{A}^{O, U}]. \quad (14)$$

Then, there exists a constant $c > 0$ such that, for any T -query algorithm $\mathcal{A}$,

$$\left| \mathbb{E}_{U \leftarrow \mathcal{D}^N} [\text{Adv}(\mathcal{A}, U)] \right| \leq \frac{cT^2}{N}. \quad (15)$$

Here, $U \leftarrow \mathcal{D}^N$ denotes that each U_k is independently sampled from $\mathcal{D}$.

We will use the following process tomography algorithm.

Theorem 2.7 ([HKOT23]). *There exists a quantum algorithm $\mathcal{A}$ that, given a black-box access to $Z \in \mathbb{U}(D)$, satisfies the following:*

- *Accuracy: On input $\epsilon, \mu \in (0, 1)$, $\mathcal{A}$ outputs a classical description of a unitary Z' such that*

$$\Pr_{Z' \leftarrow \mathcal{A}} [\|Z(\cdot)Z^\dagger - Z'(\cdot)Z'^\dagger\|_\diamond \leq \epsilon] \geq 1 - \eta. \quad (16)$$

- *Query complexity: $\mathcal{A}$ makes $O(\frac{D^2}{\epsilon} \log \frac{1}{\eta})$ queries to Z .*
- *Time complexity: The time complexity of $\mathcal{A}$ is $\text{poly}(D, \frac{1}{\epsilon}, \log \frac{1}{\eta})$.*

Lemma 2.8 (Borel-Cantelli). *Suppose that $\{X_n\}_{n \in \mathbb{N}}$ is a sequence of random variables such that $X_n \in \{0, 1\}$. If $\sum_{n \in \mathbb{N}} \mathbb{E}[X_n]$ is finite, then*

$$\Pr \left[\sum_{n \in \mathbb{N}} X_n = \infty \right] = 0. \quad (17)$$

2.3 The Haar Measure

We use the properties of the Haar measure. Recall that μ_d is the Haar measure over $\mathbb{U}(d)$, and σ_d is that over $\mathbb{S}(d)$. We will use the following concentration property.

¹⁰In [Kre21], he showed Lemma 2.6 only for the case when $\mathcal{D}$ is the Haar measure. However, his proof does not rely on any property of the Haar measure because its proof essentially depends on the BBBV theorem [BBBV97]. Thus, we can obtain the same claim for a general distribution $\mathcal{D}$ from his original proof.

Theorem 2.9 (Theorem 5.17 in [Mec19]). Given $d_1, \dots, d_k \in \mathbb{N}$, let $X = \mathbb{U}(d_1) \times \dots \times \mathbb{U}(d_k)$. Let $\mu = \mu_{d_1} \times \dots \times \mu_{d_k}$ be the product of Haar measures on X . Suppose that $f : X \rightarrow \mathbb{R}$ is L -Lipschitz in the ℓ^2 -sum of Frobenius norm, i.e., for any $U = (U_1, \dots, U_k) \in X$ and $V = (V_1, \dots, V_k) \in X$, we have $|f(U) - f(V)| \leq L \sqrt{\sum_i \|U_i - V_i\|_2^2}$. Then for every $\delta > 0$,

$$\Pr_{U \leftarrow \mu} \left[f(U) \geq \mathbb{E}_{V \leftarrow \mu} [f(V)] + \delta \right] \leq \exp \left(-\frac{(d-2)\delta^2}{24L^2} \right), \quad (18)$$

where $d := \min\{d_1, \dots, d_k\}$.

The following is a well-known fact [Har13, Mel24].

Lemma 2.10. Let $\ell, d \in \mathbb{N}$. Then,

$$\mathbb{E}_{|\psi\rangle \leftarrow \sigma_d} |\psi\rangle\langle\psi|^{\otimes \ell} = \frac{\Pi_{\text{sym}}}{\binom{d+\ell-1}{\ell}}. \quad (19)$$

Here, Π_{sym} is the following projection:

$$\Pi_{\text{sym}} = \frac{1}{\ell!} \sum_{\pi \in S_\ell} R_\pi, \quad (20)$$

where S_ℓ denotes the permutation group over ℓ elements, and R_π is permutation unitary such that $R_\pi |x_1, \dots, x_k\rangle = |x_{\pi^{-1}(1)}, \dots, x_{\pi^{-1}(k)}\rangle$ for all $x_1, \dots, x_k \in [d]$ and $\pi \in S_\ell$.

We often use the expectation of unitaries' action over some distribution. Thus, we define the following for the notational simplicity.

Definition 2.11. Let $\ell, d \in \mathbb{N}$. For a distribution ν over $\mathbb{U}(d)$, we define

$$\mathcal{M}_{\nu, \ell}(\cdot) := \mathbb{E}_{U \leftarrow \nu} U^{\otimes \ell}(\cdot) U^{\dagger \otimes \ell}. \quad (21)$$

We use a relationship between Haar random states and Haar random Choi–Jamiołkowski states.

Lemma 2.12 (Implicitly Shown in [Har23]). Let $\ell, d \in \mathbb{N}$ such that $d \geq \ell^2$. Then,

$$\left\| (\mathcal{M}_{\mu_d, \ell} \otimes \text{id})(|\Omega_{d^\ell}\rangle\langle\Omega_{d^\ell}|) - \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{d^2}} |\psi\rangle\langle\psi|^{\otimes \ell} \right\|_1 \leq O\left(\frac{\ell^2}{d}\right) \quad (22)$$

where $|\Omega_{d^\ell}\rangle = \frac{1}{d^{\ell/2}} \sum_{x \in [d]^\ell} |x\rangle |x\rangle$ is the maximally entangled state.

2.4 Unitary Complexity

For the separation, we use unitary complexity classes. First, we remind the definition of **UnitaryPSPACE**.¹¹

¹¹Note that in [BEM⁺23], Equation (24) is replaced with $\bigcap_{p \in \text{poly}} \text{UnitaryPSPACE}_{1/p}$.

Definition 2.13 (UnitaryPSPACE [RY22]). Let $\delta : \mathbb{N} \rightarrow \mathbb{R}$ be a function. We define $\text{UnitaryPSPACE}_\delta$ to be the set of a sequence of unitaries $\{U_n\}_{n \in \mathbb{N}}$ such that $\{U_n\}_{n \in \mathbb{N}}$ is quantum polynomial-space implementable¹² with error δ . Namely, there exists a quantum polynomial-space algorithm $C(\cdot, \cdot)$ such that

$$\|C(1^n, \cdot) - U_n(\cdot)U_n^\dagger\|_\diamond \leq \delta(n) \quad (23)$$

for all sufficiently large $n \in \mathbb{N}$. We also define

$$\text{UnitaryPSPACE} := \bigcap_{p \in \text{poly}} \text{UnitaryPSPACE}_{2^{-p}}. \quad (24)$$

In the definition of **UnitaryPSPACE**, intermediate measurements are allowed. [MY23] introduced a variant of **UnitaryPSPACE**, which is called **pureUnitaryPSPACE** where intermediate measurements are not allowed.¹³

Definition 2.14 (pureUnitaryPSPACE [MY23]). Let $\delta : \mathbb{N} \rightarrow \mathbb{R}$ be a function. We define $\text{pureUnitaryPSPACE}_\delta$ to be the set of a sequence of unitaries $\{U_n\}_{n \in \mathbb{N}}$ such that $\{U_n\}_{n \in \mathbb{N}}$ is quantum polynomial-space implementable with error δ and without any intermediate measurement. Namely, there exists a polynomial-space family of unitary circuit $\{C_n\}_n$ such that

$$\left\| C_n |\psi\rangle |0\dots 0\rangle - (U_n |\psi\rangle) |0\dots 0\rangle \right\| \leq \delta(n) \quad (25)$$

for all sufficiently large $n \in \mathbb{N}$ and all pure state $|\psi\rangle$. Here, $|0\dots 0\rangle$ is a state on the ancilla register. We also define

$$\text{pureUnitaryPSPACE} := \bigcap_{p \in \text{poly}} \text{pureUnitaryPSPACE}_{2^{-p}}. \quad (26)$$

Remark 2.15. It is clear that if $\{U_n\}_{n \in \mathbb{N}} \in \text{pureUnitaryPSPACE}$, we have $\{c-U_n\}_{n \in \mathbb{N}} \in \text{pureUnitaryPSPACE}$ and $\{U_n^\dagger\}_{n \in \mathbb{N}} \in \text{pureUnitaryPSPACE}$.

In [BEM⁺23], they showed that a certain problem, **SUCCINCTUHLMANN**, is **UnitaryPSPACE**-complete. For this paper, we need only the fact that **UnitaryPSPACE** has a complete problem, which is formalized as follows.¹⁴

Lemma 2.16 ([BEM⁺23]). **UnitaryPSPACE** has a complete problem. Namely, there exists a sequence of unitaries $\mathcal{U} = \{U_n\}_n$ that satisfies the following.

- $\mathcal{U} \in \text{UnitaryPSPACE}$. Moreover, $\mathcal{U} \in \text{pureUnitaryPSPACE}$.
- For any polynomial p and $\mathcal{V} = \{V_n\}_n \in \text{UnitaryPSPACE}$, there exists a QPT algorithm $\mathcal{A}^{(\cdot)}$ such that

$$\|\mathcal{A}^{\mathcal{U}}(1^n, \cdot) - V_n(\cdot)V_n^\dagger\|_\diamond \leq 2^{-p(n)} \quad (27)$$

for all sufficiently large $n \in \mathbb{N}$.

¹²Here, intermediate measurements are allowed.

¹³It is trivial that $\text{pureUnitaryPSPACE} \subseteq \text{UnitaryPSPACE}$. However, the other direction is not trivial, because a quantum polynomial-space algorithm C that implements U_n with an exponentially small error could perform exponentially many intermediate measurements, but postponing these measurements requires exponentially many ancilla qubits.

¹⁴There are two remarks. First, [BEM⁺23] showed that **SUCCINCTUHLMANN** is in **UnitaryPSPACE**. However, their proof can be easily modified so that the problem is in **pureUnitaryPSPACE**. Second, in [BEM⁺23], their definition of the reduction and **UnitaryPSPACE** allows the inverse polynomial error, while we only allow the exponentially small error in Equation (27) and Equation (24). However, we can easily see their original proof of Theorem 7.14 [BEM⁺23] also works in our case.

2.5 Quantum Singular Value Transformation and Block Encoding

To break PRUs, we use a quantum singular value transformation (QSVT) [GSLW19]. Especially, we use the following singular value discrimination algorithm.

Theorem 2.17 (Singular Value Discrimination Algorithm [GSLW19]). *Let $0 \leq a < b \leq 1$. Suppose that $M \in L(d)$ can be written $M = \tilde{\Pi}U\Pi$ with some $U \in \mathbb{U}(d)$ and projections $\Pi, \tilde{\Pi} \in L(d)$. Let ξ be a given unknown state promised that*

- *the support of ξ is contained in the subspace W_0 , which is the subspace spanned by the right singular vectors of M with singular value at most a or*
- *the support of ξ is contained in the subspace W_1 , which is the subspace spanned by the right singular vectors of M with singular value at least b .*

Then, for each $\eta > 0$, there exists an algorithm $\mathcal{D}$ satisfying the following:

- *on input a single copy of ξ , $\mathcal{D}$ distinguishes between the first case or the second case with probability at least $1 - \eta$;*
- *$\mathcal{D}$ uses $U, U^\dagger, C_\Pi \text{NOT}, C_{\tilde{\Pi}} \text{NOT}$ and other single-qubit gates*

$$O\left(\frac{1}{\max\{b-a, \sqrt{1-a^2}-\sqrt{1-b^2}\}} \log\left(\frac{1}{\eta}\right)\right)$$

times, and uses a single ancilla qubit. Here, $C_\Pi \text{NOT} := \Pi \otimes X + (I - \Pi) \otimes I$ and $C_{\tilde{\Pi}} \text{NOT}$ is defined in the same way.

When we apply the singular value discrimination algorithm, we need to encode a matrix into a unitary circuit. This technique is referred to as block encoding.

Definition 2.18 (Block Encoding [GSLW19]). *Let $M \in L(d)$. We say that $U \in \mathbb{U}(2^a d)$ is an (α, ϵ, a) -block encoding of M for some $\alpha \geq 1, \epsilon \geq 0$ and $a \in \mathbb{N}$ if it satisfies*

$$\|M - \alpha(|0^a\rangle\langle 0^a| \otimes I)U(|0^a\rangle\langle 0^a| \otimes I)\|_\infty \leq \epsilon. \quad (28)$$

In general, it is not clear that we can space-efficiently implement a block encoding unitary of any matrix M . The following lemma ensures that we can implement a block encoding unitary of a density operator if we can generate its purification.

Lemma 2.19 (Lemma 12 in [vAG19]). *Let U be a unitary over registers $\mathbf{A}$ and $\mathbf{B}$, where $\mathbf{A}$ and $\mathbf{B}$ are n -qubit register and m -qubit register, respectively. Define $\rho_{\mathbf{A}} := \text{Tr}_{\mathbf{B}}[(U|0\dots 0\rangle\langle 0\dots 0|U^\dagger)_{\mathbf{AB}}]$. Then, there exists a $(1, 0, n+m)$ -block encoding unitary V of $\rho_{\mathbf{A}}$, where V is implementable with single use of U and $U^\dagger$, and $n+1$ two-qubit gates.*

2.6 Cryptographic Primitives

We recall PRUs defined by [JLS18].

Definition 2.20 (Pseudorandom Unitaries [JLS18]). *We define that an algorithm G is a pseudorandom unitary generator (PRU) if it satisfies the following:*

- *Correctness:* Let λ be the security parameter. Let $\mathcal{K}_\lambda$ denote the key-space at most $\text{poly}(\lambda)$ bits. G is a QPT algorithm such that $G(k, |\psi\rangle) = U_k |\psi\rangle$ for any λ -qubit state $|\psi\rangle$.
- *Pseudorandomness:* For any uniform QPT algorithm $\mathcal{A}^{(\cdot)}$,

$$\left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{U_k}(1^\lambda)] - \Pr_{U \leftarrow \mu_{2^\lambda}} [1 \leftarrow \mathcal{A}^U(1^\lambda)] \right| \leq \text{negl}(\lambda). \quad (29)$$

If Equation (29) holds for any non-adaptively-querying adversary, we call G non-adaptive PRU.¹⁵ If $G(k, \cdot)$ uses at most c ancilla qubits to implement U_k for all $k \in \mathcal{K}_\lambda$, we call G a c -ancilla PRU.

Remark 2.21. We could define PRUs secure against non-uniform adversaries, but in this paper we can break PRUs against uniform adversaries, and therefore we provide only the definition of the latter.

Definition 2.22 (Pseudorandom Isometries [AGKL24]). Let $s : \mathbb{N} \mapsto \mathbb{N}$ be a function such that $s(\lambda) \leq \text{poly}(\lambda)$. We define that an algorithm G is a pseudorandom isometry generator with s stretch (PRI) if it satisfies the following:

- *Correctness:* Let λ be the security parameter. Let $\mathcal{K}_\lambda$ denote the key-space at most $\text{poly}(\lambda)$ qubits. G is a QPT algorithm such that $G(k, |\psi\rangle) = \mathcal{I}_k |\psi\rangle$ for any λ -qubit state $|\psi\rangle$, where $\mathcal{I}_k$ is an isometry that maps λ qubits to $\lambda + s(\lambda)$ qubits.
- *Pseudorandomness:* For any uniform QPT algorithm $\mathcal{A}^{(\cdot)}$,

$$\left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{I}_k}(1^\lambda)] - \Pr_{U \leftarrow \mu_{2^{\lambda+s(\lambda)}}} [1 \leftarrow \mathcal{A}^{\mathcal{I}_U}(1^\lambda)] \right| \leq \text{negl}(\lambda), \quad (30)$$

where, for each $U \in \mathbb{U}(2^{\lambda+s(\lambda)})$, $\mathcal{I}_U$ is the isometry that maps λ -qubit state $|\psi\rangle$ to $(\lambda + s(\lambda))$ -qubit state $U(|\psi\rangle |0^s\rangle)$.¹⁶

If Equation (30) holds for any non-adaptively-querying adversary, we call G non-adaptive PRI with s stretch.¹⁷ If $G(k, \cdot)$ uses at most $s + c$ ancilla qubits to implement $\mathcal{I}_k$ for all $k \in \mathcal{K}_\lambda$, we call G a c -ancilla PRI with s stretch.¹⁸

Quantumly-accessible adaptively-secure PRFSGs were defined in [AGQY22].

Definition 2.23 (Quantumly-accessible adaptively-secure PRFSGs [AGQY22]). We define that an algorithm G is a quantumly-accessible adaptively-secure PRFSG if it satisfies the following:

- *Correctness:* Let $\lambda \in \mathbb{N}$ be the security parameter. Let q be a polynomial. Let $\mathcal{K}_\lambda$ denote the key-space at most $\text{poly}(\lambda)$ bits. G is a QPT algorithm that takes a key $k \in \mathcal{K}_\lambda$ and a bit string x as input, and outputs a pure $q(\lambda)$ -qubit state $|\phi_k(x)\rangle$.

¹⁵Here, non-adaptive query means that the adversary queries $U^{\otimes \text{poly}(\lambda)}$ only once.

¹⁶Without loss of generality, the ancilla state can be $|0^s\rangle$ due to the right invariance of the Haar measure.

¹⁷Here, non-adaptive query means that the adversary queries $\mathcal{I}^{\otimes \text{poly}(\lambda)}$ only once.

¹⁸To implement an isometry from λ qubits to $(\lambda + s)$ qubits, we need at least s ancilla qubits.

- *Quantumly-accessible adaptive security:* For any QPT adversary $\mathcal{A}^{(\cdot)}$ and any bit string y whose length is at most polynomial of λ ,

$$\left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{G(k, \cdot)}(1^\lambda, y)] - \Pr_{\{|\vartheta_x\rangle\} \leftarrow \sigma} [1 \leftarrow \mathcal{A}^{\mathcal{H}_{\{|\vartheta_x\rangle\}}}(1^\lambda, y)] \right| \leq \text{negl}(\lambda), \quad (31)$$

where $\{|\vartheta_x\rangle\} \leftarrow \sigma$ denotes that each $|\vartheta_x\rangle$ is independently chosen from the Haar measure σ_{2^q} . Here, the actions of $G(k, \cdot)$ and $\mathcal{H}_{\{|\vartheta_x\rangle\}}$ are defined as follows:

- $G(k, \cdot)$: It applies $|x\rangle \mapsto |x\rangle |\phi_k(x)\rangle$ coherently.¹⁹
- $\mathcal{H}_{\{|\vartheta_x\rangle\}}$: It applies $|x\rangle \mapsto |x\rangle |\vartheta_x\rangle$ coherently.

In this paper, we often omit the term "quantumly-accessible adaptively-secure".

Remark 2.24. Here we provide the definition of PRFSGs secure against non-uniform adversaries with classical advice because we can construct it. We can also consider the security against all non-uniform adversaries with quantum advice, but it is not clear whether our construction Definition 4.1 satisfies the security. We leave it to the future work.

3 Separation Oracle

In this section, we define an oracle that separates between PRUs and PRFSGs. Our separation oracle is defined as follows:

Definition 3.1 (Separation Oracle). We define an oracle $\mathcal{O} := (S, \mathcal{U})$ as follows:

- For each $n \in \mathbb{N}$ and $m \in \{0, 1\}^n$, sample $|\psi_{n,m}\rangle \in \mathbb{S}(2^n)$ from the Haar measure σ_{2^n} . Then, define the $(n+1)$ -qubit swapping unitary

$$\mathcal{S}_{n,m} := |0\rangle\langle 1| \otimes |0^n\rangle\langle \psi_{n,m}| + |1\rangle\langle 0| \otimes |\psi_{n,m}\rangle\langle 0^n| + I_\perp^{n,m} \quad (32)$$

for each $n \in \mathbb{N}$ and $m \in \{0, 1\}^n$. Here, $I_\perp^{n,m}$ is the identity on the subspace orthogonal to $\text{span}\{|0\rangle\langle 0^n|, |1\rangle\langle \psi_{n,m}|\}$. We define $\mathcal{S} := \{\mathcal{S}_n\}_{n \in \mathbb{N}}$, where $\mathcal{S}_n := \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes \mathcal{S}_{n,m}$ is a $(2n+1)$ -qubit unitary.

- $\mathcal{U} := \{U_n\}_{n \in \mathbb{N}}$ is the **UnitaryPSPACE** complete problem in Lemma 2.16.

In this work, we allow not only the query to $\mathcal{O}$ but also the query to the inverse of $\mathcal{O}$. When we write $\mathcal{A}^{\mathcal{O}}$ for an algorithm $\mathcal{A}$, it can query $\mathcal{O}$ and the inverse of $\mathcal{O}$.

Remark 3.2. For any $|\psi_{n,m}\rangle \in \mathbb{S}(2^n)$, $\mathcal{S}_{n,m} = \mathcal{S}_{n,m}^\dagger$ by its definition. Thus, when we consider an algorithm $\mathcal{A}^{\mathcal{S}}$, it suffices to consider the forward query to $\mathcal{S}$ regardless of the choice of $\mathcal{S}$.

Our goal is to show the following.

Theorem 3.3. With probability 1 over the choice of $\mathcal{O}$ defined in Definition 3.1, the following are satisfied:

1. Quantumly-accessible adaptively-secure PRFSGs exist relative to $\mathcal{O}$.
2. Non-adaptive, $O(\log \lambda)$ -ancilla PRUs do not exist relative to $\mathcal{O}$.

We prove the existence of PRFSGs in Section 4, and the non-existence of PRUs in Section 5.

¹⁹When a superposition $\sum_x \alpha_x |x\rangle |\xi_x\rangle$ is queried, it outputs $\sum_x \alpha_x |x\rangle |\phi_k(x)\rangle |\xi_x\rangle$. In general it is not possible when the junk states depending on x appear, but in our construction of PRFSG, junk states are independent of x .

4 Constructing PRFSGs

In this section, we show that quantumly-accessible adaptively-secure PRFSGs exist relative to $\mathcal{O}$. We construct PRFSGs as follows:

Definition 4.1. Let $\mathcal{O} = (\mathcal{S}, \mathcal{U})$ be the oracle in Definition 3.1. Relative to $\mathcal{O}$, we define a QPT algorithm $G^{\mathcal{O}}$ as follows:

1. Let $k, x \in \{0, 1\}^\lambda$ be an input.²⁰ Here, k is a secret key and x is an input bit string.
2. Prepare $|(k, x)\rangle |0\rangle |0^{2\lambda}\rangle$. Here (k, x) denotes the concatenation of k and x .
3. Obtain $|(k, x)\rangle |1\rangle |\psi_{2\lambda, (k, x)}\rangle$ by querying $|(k, x)\rangle |0\rangle |0^{2\lambda}\rangle$ to $\mathcal{S}_{2\lambda}$.
4. Output $|\psi_{2\lambda, (k, x)}\rangle$.

The goal of this section is to prove the following:

Theorem 4.2. With probability 1 over the randomness of $\mathcal{O}$ (defined in Definition 3.1), Definition 4.1 is a quantumly-accessible adaptively-secure PRFSGs relative to $\mathcal{O}$.

Our strategy is the same as [Kre21] which shows PRUs exist relative to exponentially many Haar random unitary oracles. As a first step, we need that swap oracles are indistinguishable from independent swap oracles on average. This is formalized as follows and directly follows from Lemma 2.6.

Lemma 4.3. Let $\mathcal{A}^{(\cdot, \cdot)}$ be an algorithm. Let $\lambda \in \mathbb{N}$. For each fixed $\mathcal{S}$ defined in Definition 3.1, we define

$$\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda}) := \Pr_{k \leftarrow \{0, 1\}^\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{2\lambda, k}, \mathcal{S}_{2\lambda}}] - \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2\lambda}} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \mathcal{S}_{2\lambda}}], \quad (33)$$

where

- $\mathcal{T}_{2\lambda, k} := \sum_{x \in \{0, 1\}^\lambda} |x\rangle\langle x| \otimes \mathcal{S}_{2\lambda, (k, x)}$.
- for $|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle$, $\mathcal{T}_{\{|\vartheta_x\rangle\}} := \sum_{x \in \{0, 1\}^\lambda} |x\rangle\langle x| \otimes \mathcal{T}_{|\vartheta_x\rangle}$. Here, for $|\vartheta\rangle \in \mathbb{S}(2^{2\lambda})$, we define $\mathcal{T}_{|\vartheta\rangle} := |0\rangle\langle 1| \otimes |0^{2\lambda}\rangle\langle \vartheta| + |1\rangle\langle 0| \otimes |\vartheta\rangle\langle 0^{2\lambda}| + I_\perp^{|\vartheta\rangle}$, where $I_\perp^{|\vartheta\rangle}$ is the identity on the subspace orthogonal to $\text{span}\{|0\rangle |0^{2\lambda}\rangle, |1\rangle |\vartheta\rangle\}$.

Then, there exists a constant $c > 0$ such that, for any algorithm $\mathcal{A}^{(\cdot, \cdot)}$ that makes T queries in total,

$$\left| \mathbb{E}_{\mathcal{S}_{2\lambda} \leftarrow \sigma} [\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda})] \right| \leq \frac{cT^2}{2^\lambda}, \quad (34)$$

where $\mathcal{S}_{2\lambda} \leftarrow \sigma$ denotes that, for each $m \in \{0, 1\}^{2\lambda}$, $|\psi_{2\lambda, m}\rangle$ is drawn from the Haar measure $\sigma_{2\lambda}$ independently.

Proof of Lemma 4.3. Note that $\mathcal{S}_{2\lambda} = \sum_{k \in \{0, 1\}^\lambda} |k\rangle\langle k| \otimes \mathcal{T}_{2\lambda, k}$. Thus, this claim follows from Lemma 2.6 with $N = 2^\lambda$, and $\mathcal{D} = \sigma$. $\square$

²⁰In our explicit construction, the length of the secret key is the same as that of the input bit string. However, our security proof works if they are different. On the other hand, the number of qubits of output states must be larger than the length of the secret key and input bit string. In particular, it is not clear whether short PRFSGs exist or not relative to our oracle. We leave it to further work.

Next, we want to show that $\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda})$ is negligible with overwhelming probability over the choice of $\mathcal{S}_{2\lambda}$ by invoking the concentration inequality (Theorem 2.9). For that goal, we view $\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda})$ as a function of $U \in \mathbb{U}(2^{2\lambda})^{2^{2\lambda}}$, and need to show that it satisfies Lipschitz condition in Theorem 2.9. This is formalized as follows:

Lemma 4.4. *Let $n \in \mathbb{N}$. For $U = (U_1, \dots, U_{2^n}) \in \mathbb{U}(2^n)^{2^n}$, we define*

$$\tilde{\mathcal{S}}_n(U) := \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes \left(|0\rangle\langle 1| \otimes |0^n\rangle\langle 0^n| U_m^\dagger + |1\rangle\langle 0| \otimes U_m |0^n\rangle\langle 0^n| + I_\perp(U_m) \right), \quad (35)$$

where each $I_\perp(U_m)$ is the identity on the subspace orthogonal to $\text{span}\{|0\rangle|0^n\rangle, |1\rangle U_m |0^n\rangle\}$. Then, for any algorithm $\mathcal{A}^{(\cdot)}$ that makes T queries, $f(U) := \Pr[1 \leftarrow \mathcal{A}^{\tilde{\mathcal{S}}_n(U)}]$ is $8T$ -Lipschitz in the ℓ^2 -sum of the Forbenious norm. Namely, for any $U = (U_1, \dots, U_{2^n}), V = (V_1, \dots, V_{2^n}) \in \mathbb{U}(2^n)^{2^n}$,

$$|f(U) - f(V)| \leq 8T \sqrt{\sum_{m \in \{0,1\}^n} \|U_m - V_m\|_2^2}. \quad (36)$$

Proof of Lemma 4.4. Note that, for each $m \in \{0,1\}^n$, we can write $I_\perp(U_m)$ as follows:

$$I_\perp(U_m) = I - |0\rangle\langle 0| \otimes |0^n\rangle\langle 0^n| - |1\rangle\langle 1| \otimes U_m |0^n\rangle\langle 0^n| U_m^\dagger. \quad (37)$$

Thus, we have

$$\begin{aligned} & |f(U) - f(V)| \\ & \leq 2T \|\tilde{\mathcal{S}}_n(U) - \tilde{\mathcal{S}}_n(V)\|_2 \quad (\text{By Lemma 2.3}) \\ & = 2T \left\| \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes \left(|0\rangle\langle 1| \otimes |0^n\rangle\langle 0^n| (U_m - V_m)^\dagger + |1\rangle\langle 0| \otimes (U_m - V_m) |0^n\rangle\langle 0^n| \right. \right. \\ & \quad \left. \left. - |1\rangle\langle 1| \otimes U_m |0^n\rangle\langle 0^n| U_m + |1\rangle\langle 1| \otimes V_m |0^n\rangle\langle 0^n| V_m^\dagger \right) \right\|_2 \quad (\text{By Equation (37)}) \\ & \leq 2T \left\| \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes \left(|0\rangle\langle 1| \otimes |0^n\rangle\langle 0^n| (U_m - V_m)^\dagger + |1\rangle\langle 0| \otimes (U_m - V_m) |0^n\rangle\langle 0^n| \right) \right\|_2 \\ & \quad + 2T \left\| \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes |1\rangle\langle 1| \otimes \left(U_m |0^n\rangle\langle 0^n| U_m - V_m |0^n\rangle\langle 0^n| V_m^\dagger \right) \right\|_2, \quad (38) \end{aligned}$$

where the last inequality follows from the triangle inequality. The first term in Equation (38) is estimated as follows:

$$\begin{aligned}
& 2T \left\| \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes \left(|0\rangle\langle 1| \otimes |0^n\rangle\langle 0^n| (U_m - V_m)^\dagger + |1\rangle\langle 0| \otimes (U_m - V_m) |0^n\rangle\langle 0^n| \right) \right\|_2 \\
& \leq 4T \left\| \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes |1\rangle\langle 0| \otimes (U_m - V_m) |0^n\rangle\langle 0^n| \right\|_2 \quad (\text{By } \|A^\dagger\|_2 = \|A\|_2 \text{ and the triangle inequality}) \\
& = 4T \sqrt{\sum_{m \in \{0,1\}^n} \left\| |1\rangle\langle 0| \otimes (U_m - V_m) |0^n\rangle\langle 0^n| \right\|_2^2} \quad (\text{By } \|\sum_m |m\rangle\langle m| \otimes A_m\|_2 = \sqrt{\sum_m \|A_m\|_2^2}) \\
& = 4T \sqrt{\sum_{m \in \{0,1\}^n} \left\| (U_m - V_m) |0^n\rangle\langle 0^n| \right\|_2^2} \quad (\text{By } \|A \otimes B\|_2 = \|A\|_2 \|B\|_2 \text{ and } \| |1\rangle\langle 0| \|_2 = 1) \\
& \leq 4T \sqrt{\sum_{m \in \{0,1\}^n} \|U_m - V_m\|_2^2}, \tag{39}
\end{aligned}$$

where the last inequality follows from the Hölder's inequality (Lemma 2.2) and $\| |0^n\rangle\langle 0^n| \|_\infty = 1$. On the other hand, the second term in Equation (38) is estimated as follows:

$$\begin{aligned}
& 2T \left\| \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes |1\rangle\langle 1| \otimes \left(U_m |0^n\rangle\langle 0^n| U_m - V_m |0^n\rangle\langle 0^n| V_m^\dagger \right) \right\|_2 \\
& = 2T \sqrt{\sum_{m \in \{0,1\}^n} \left\| U_m |0^n\rangle\langle 0^n| U_m - V_m |0^n\rangle\langle 0^n| V_m^\dagger \right\|_2^2} \quad (\text{By } \|\sum_m |m\rangle\langle m| \otimes A_m\|_2 = \sqrt{\sum_m \|A_m\|_2^2}) \\
& \leq 4T \sqrt{\sum_{m \in \{0,1\}^n} \|U_m - V_m\|_2^2}, \tag{40}
\end{aligned}$$

where the last inequality follows from Lemma 2.5. By combining Equations (38) to (40), we have

$$|f(U) - f(V)| \leq 8T \sqrt{\sum_{m \in \{0,1\}^n} \|U_m - V_m\|_2^2}, \tag{41}$$

which concludes the proof. $\square$

With Lemmata 4.3 and 4.4 at hand, we can argue $\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda})$ is negligible with high probability.

Lemma 4.5. *Let c be a constant in Lemma 4.3. Suppose that $\mathcal{A}^{(\cdot, \cdot)}$ is an algorithm that makes T queries in total. Then, for any $p \geq \frac{cT^2}{2^\lambda}$,*

$$\Pr_{S_{2\lambda} \leftarrow \sigma} [|\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda})| \geq p] \leq 2 \exp \left(- \frac{(2^{2\lambda} - 2)(p - cT^2 2^{-\lambda})^2}{6144T^2} \right), \tag{42}$$

where $\text{Adv}(\mathcal{A}, \mathcal{S}_{2\lambda})$ is defined in Lemma 4.3, and $S_{2\lambda} \leftarrow \sigma$ denotes that, for each $m \in \{0,1\}^{2\lambda}$, $|\psi_{2\lambda,m}\rangle$ is drawn from the Haar measure $\sigma_{2^{2\lambda}}$ independently.

Proof of Lemma 4.5. Recall that $S_{2\lambda} = \sum_{m \in \{0,1\}^{2\lambda}} |m\rangle\langle m| \otimes S_{2\lambda,m}$ and each $S_{2\lambda,m}$ is a unitary that swaps between $|0\rangle |0^{2\lambda}\rangle$ and $|1\rangle |\psi_{2\lambda,m}\rangle$. It is clear that choosing $|\psi_{2\lambda,m}\rangle$ from the Haar measure $\sigma_{2^{2\lambda}}$ independently

is exactly the same as setting $|\psi_{2\lambda,m}\rangle := U_m |0^{2\lambda}\rangle$, where U_m is chosen from the Haar measure $\mu_{2^{2\lambda}}$ independently. Thus, let $\tilde{\mathcal{S}}_{2\lambda} \leftarrow \mu$ denote that, for each $m \in \{0, 1\}^{2\lambda}$, $|\psi_{2\lambda,m}\rangle$ is defined by $U_m |0^{2\lambda}\rangle$ and $U_m \leftarrow \mu_{2^{2\lambda}}$. Recall that

$$\text{Adv}(\mathcal{A}, \tilde{\mathcal{S}}_{2\lambda}) = \Pr_{k \leftarrow \{0,1\}^\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{2\lambda,k}, \tilde{\mathcal{S}}_{2\lambda}}] - \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2^{2\lambda}}} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \tilde{\mathcal{S}}_{2\lambda}}]. \quad (43)$$

We can see that both teams are $8T$ -Lipschitz functions as follows:

- $\Pr_{k \leftarrow \{0,1\}^\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{2\lambda,k}, \tilde{\mathcal{S}}_{2\lambda}}]$: define the following algorithm $\mathcal{A}_1^{\tilde{\mathcal{S}}_{2\lambda}}$.
 1. Choose $k \leftarrow \{0, 1\}^\lambda$.
 2. Simulate $\mathcal{A}^{\mathcal{T}_{2\lambda,k}, \tilde{\mathcal{S}}_{2\lambda}}$ with the query access to $\tilde{\mathcal{S}}_{2\lambda}$. When $\mathcal{A}$ queries a register $\mathbf{A}$ to $\mathcal{T}_{2\lambda,k}$, simulate its query by preparing $|k\rangle$ on an ancilla register $\mathbf{R}$ and querying $\mathbf{RA}$ to $\tilde{\mathcal{S}}_{2\lambda}$.
 3. Output what $\mathcal{A}$ outputs.

It is clear that $\Pr[1 \leftarrow \mathcal{A}_1^{\tilde{\mathcal{S}}_{2\lambda}}] = \Pr_{k \leftarrow \{0,1\}^\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{2\lambda,k}, \tilde{\mathcal{S}}_{2\lambda}}]$, and $\mathcal{A}_1^{\tilde{\mathcal{S}}_{2\lambda}}$ makes T queries. Thus, from Lemma 4.4, we can view that $\Pr_{k \leftarrow \{0,1\}^\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{2\lambda,k}, \tilde{\mathcal{S}}_{2\lambda}}]$ is an $8T$ -Lipschitz function in the ℓ^2 -sum of Frobenious norm.

- $\Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2^{2\lambda}}} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \tilde{\mathcal{S}}_{2\lambda}}]$: define the following algorithm $\mathcal{A}_2^{\tilde{\mathcal{S}}_{2\lambda}}$.
 1. Choose $|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2^{2\lambda}}$ along with their classical descriptions.
 2. Simulate $\mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \tilde{\mathcal{S}}_{2\lambda}}$. When $\mathcal{A}$ queries $\mathcal{T}_{\{|\vartheta_x\rangle\}}$, $\mathcal{A}_2$ applies $\mathcal{T}_{\{|\vartheta_x\rangle\}}$ by using the classical descriptions of $|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle$.
 3. Output what $\mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \tilde{\mathcal{S}}_{2\lambda}}$ outputs.

It is clear that $\Pr[1 \leftarrow \mathcal{A}_2^{\tilde{\mathcal{S}}_{2\lambda}}] = \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2^{2\lambda}}} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \tilde{\mathcal{S}}_{2\lambda}}]$, and $\mathcal{A}_2^{\tilde{\mathcal{S}}_{2\lambda}}$ makes T queries. Thus, from Lemma 4.4, we can view that $\Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2^{2\lambda}}} [1 \leftarrow \mathcal{A}^{\mathcal{T}_{\{|\vartheta_x\rangle\}}, \tilde{\mathcal{S}}_{2\lambda}}]$ is an $8T$ -Lipschitz function in the ℓ^2 -sum of Frobenious norm.

Thus, from Lemma 2.4, we can view $\text{Adv}(\mathcal{A}, \tilde{\mathcal{S}}_{2\lambda})$ is a function that maps $\mathbb{U}(2^{2\lambda})^{2^{2\lambda}} \rightarrow \mathbb{R}$ and it is a $16T$ -Lipschitz in the ℓ^2 -sum of Frobenious norm. Therefore,

$$\begin{aligned} & \Pr_{\tilde{\mathcal{S}}_{2\lambda} \leftarrow \mu} [\text{Adv}(\mathcal{A}, \tilde{\mathcal{S}}_{2\lambda}) \geq p] \\ & \leq \Pr_{\tilde{\mathcal{S}}_{2\lambda} \leftarrow \mu} \left[\text{Adv}(\mathcal{A}, \tilde{\mathcal{S}}_{2\lambda}) \geq \mathbb{E}_{\tilde{\mathcal{S}}'_{2\lambda} \leftarrow \mu} [\text{Adv}(\mathcal{A}, \tilde{\mathcal{S}}'_{2\lambda})] + p - cT^2 2^{-\lambda} \right] \quad (\text{By Lemma 4.3}) \\ & \leq \exp \left(- \frac{(2^{2\lambda} - 2)(p - cT^2 2^{-\lambda})^2}{6144T^2} \right). \quad (\text{By Theorem 2.9 with } d = 2^{2\lambda}, \delta = p - cT^2 2^{-\lambda}, L = 16T) \end{aligned}$$

We can obtain the same bound for $\Pr_{\tilde{\mathcal{S}}_{2\lambda} \leftarrow \mu} [\text{Adv}(\mathcal{A}, \tilde{\mathcal{S}}_{2\lambda}) \leq -p]$ in the same way, so we obtain our claim. $\square$

Now we are ready to prove Theorem 4.2. We restate it here for the reader's convenience.

Theorem 4.2. *With probability 1 over the randomness of $\mathcal{O}$ (defined in Definition 3.1), Definition 4.1 is a quantumly-accessible adaptively-secure PRFSGs relative to $\mathcal{O}$.*

Proof of Theorem 4.2. First, we recall Definition 4.1: For a fixed $\mathcal{O} = (\mathcal{S}, \mathcal{U})$ and inputs $k, x \in \{0, 1\}^\lambda$, $G^\mathcal{O}(k, x) = |\psi_{2\lambda, (k, x)}\rangle$. From Definition 4.1, it is clear G is a QPT algorithm. Thus, it suffices to show that G satisfies quantumly-accessible adaptive security.

From Remark 3.2, for the oracle $\mathcal{S}$, it suffices to consider the forward query. Recall that, for fixed $\mathcal{O}$ and $k \in \{0, 1\}^\lambda$, quantum query to $G^\mathcal{O}(k, \cdot)$ is defined as follows: let $\sum_x \alpha_x |x\rangle_{\mathbf{X}} |\xi_x\rangle_{\mathbf{Z}}$ be an overall state of the adversary $\mathcal{A}$. When $\mathcal{A}$ queries the register $\mathbf{X}$, return $\sum_x \alpha_x |x\rangle_{\mathbf{X}} |\psi_{2\lambda, (k, x)}\rangle_{\mathbf{Y}} |\xi_x\rangle_{\mathbf{Z}}$. Similarly, when $\mathcal{A}$ queries quantumly the register $\mathbf{X}$ to the ideal oracle $\mathcal{H}_{\{|\vartheta_x\rangle\}}$, it returns $\sum_x \alpha_x |x\rangle_{\mathbf{X}} |\vartheta_x\rangle_{\mathbf{Y}} |\xi_x\rangle_{\mathbf{Z}}$.

To show the security, we want to invoke Lemma 4.5. However, since the above oracles $G^\mathcal{O}(k, \cdot)$ and $\mathcal{H}_{\{|\vartheta_x\rangle\}}$ are different from $\mathcal{T}_{2\lambda, k}$ and $\mathcal{T}_{2\lambda, \{|\vartheta_x\rangle\}}$, we cannot invoke Lemma 4.5 directly. For that purpose, we construct an algorithm $\mathcal{B}^{(\cdot), S_{2\lambda}}$ with the query access to $\mathcal{T}_{2\lambda, k}$ or $\mathcal{T}_{2\lambda, \{|\vartheta_x\rangle\}}$ that simulates $\mathcal{A}^{(\cdot), S_{2\lambda}}$ with the query access to $G^\mathcal{O}(k, \cdot)$ or $\mathcal{H}_{\{|\vartheta_x\rangle\}}$, respectively:

1. $\mathcal{B}$ simulates $\mathcal{A}$. When $\mathcal{A}$ queries the first oracles, $\mathcal{B}$ simulates as follows.
 - When $\mathcal{A}$ queries the register $\mathbf{X}$ to $S_{2\lambda}$, $\mathcal{B}$ queries it to $S_{2\lambda}$.
 - When $\mathcal{A}$ queries the register $\mathbf{X}$ to the first oracle ($G^\mathcal{O}(k, \cdot)$ or $\mathcal{H}_{\{|\vartheta_x\rangle\}}$), $\mathcal{B}$ prepares $|0\rangle_{\mathbf{A}} |0^{2\lambda}\rangle_{\mathbf{Y}}$. Then, $\mathcal{B}$ queries the registers $\mathbf{X}$, $\mathbf{A}$ and $\mathbf{Y}$ to the first oracle ($\mathcal{T}_{2\lambda, k}$ or $\mathcal{T}_{2\lambda, \{|\vartheta_x\rangle\}}$), and removes the register $\mathbf{A}$.
2. $\mathcal{B}$ outputs what $\mathcal{A}$ outputs.

It is clear that

$$\Pr_{k \leftarrow \{0, 1\}^\lambda} [1 \leftarrow \mathcal{B}^{\mathcal{T}_{2\lambda, k}, S_{2\lambda}}] = \Pr_{k \leftarrow \{0, 1\}^\lambda} [1 \leftarrow \mathcal{A}^{G^\mathcal{O}(k, \cdot), S_{2\lambda}}] \quad (44)$$

and

$$\Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2\lambda}} [1 \leftarrow \mathcal{B}^{\mathcal{T}_{2\lambda, \{|\vartheta_x\rangle\}}, S_{2\lambda}}] = \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2\lambda}} [1 \leftarrow \mathcal{A}^{\mathcal{H}_{\{|\vartheta_x\rangle\}}, S_{2\lambda}}]. \quad (45)$$

Thus, for any adversary $\mathcal{A}$ with classical advice y ,

$$\begin{aligned} & \Pr_{S_{2\lambda} \leftarrow \sigma} \left[\left| \Pr_{k \leftarrow \{0, 1\}^\lambda} [1 \leftarrow \mathcal{A}^{G(k, \cdot), S_{2\lambda}}(1^\lambda, y)] - \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2\lambda}} [1 \leftarrow \mathcal{A}^{\mathcal{H}_{\{|\vartheta_x\rangle\}}, S_{2\lambda}}(1^\lambda, y)] \right| \geq \frac{1}{2^{\lambda/2}} \right] \\ &= \Pr_{S_{2\lambda} \leftarrow \sigma} \left[\left| \Pr_{k \leftarrow \{0, 1\}^\lambda} [1 \leftarrow \mathcal{B}^{\mathcal{T}_{2\lambda, k}, S_{2\lambda}}(1^\lambda, y)] - \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2\lambda}} [1 \leftarrow \mathcal{B}^{\mathcal{T}_{2\lambda, \{|\vartheta_x\rangle\}}, S_{2\lambda}}(1^\lambda, y)] \right| \geq \frac{1}{2^{\lambda/2}} \right] \\ &\leq 2 \exp \left(- \frac{(2^{2\lambda} - 2)(2^{-\lambda/2} - cT^2 2^{-\lambda})^2}{6144T^2} \right) \quad (\text{By Lemma 4.5 with } p = 2^{-\lambda/2}) \\ &\leq \exp \left(- O \left(\frac{2^\lambda}{T^2} \right) \right). \end{aligned} \quad (46)$$

Hence, for any T -query adversary $\mathcal{A}$ and any polynomial q ,

$$\begin{aligned} & \Pr_{S_{2\lambda} \leftarrow \sigma} \left[\text{there exists a } y \in \{0, 1\}^q \text{ s.t.} \right. \\ & \quad \left| \Pr_{k \leftarrow \{0, 1\}^\lambda} [1 \leftarrow \mathcal{A}^{G(k, \cdot), S_{2\lambda}}(1^\lambda, y)] - \Pr_{|\vartheta_1\rangle, \dots, |\vartheta_{2\lambda}\rangle \leftarrow \sigma_{2\lambda}} [1 \leftarrow \mathcal{A}^{\mathcal{H}_{\{|\vartheta_x\rangle\}}, S_{2\lambda}}(1^\lambda, y)] \right| \geq \frac{1}{2^{\lambda/2}} \left. \right] \\ &\leq 2^q \exp \left(- O \left(\frac{2^\lambda}{T^2} \right) \right) \quad (\text{By the union bound and Equation (46)}) \\ &\leq \text{negl}(\lambda). \end{aligned} \quad (47)$$

Since all $\mathcal{S}_n$ with $n \neq 2\lambda$ and $\mathcal{U}$ are independent of $\mathcal{S}_{2\lambda}$, the above inequality holds even if $\mathcal{A}$ queries $\mathcal{S}_n$ with $n \neq 2\lambda$ and $\mathcal{U}$. Therefore, by applying the Borel-Cantelli lemma (Lemma 2.8) with $\sum_{\lambda} \text{negl}(\lambda) \leq \sum_{\lambda} \lambda^{-2} \leq \infty$, no adversaries with classical advice can distinguish Definition 4.1 from independent Haar random states with at least $2^{-\lambda/2}$ advantage relative to $\mathcal{O}$ with probability 1 over the randomness of $\mathcal{O}$. This concludes the proof. $\square$

Remark 4.6. As in the case of [Kre21], it is not clear whether Definition 4.1 is secure even against adversaries with quantum advice or not. In [Kre21], he gives an idea to extend the security proof against adversaries with quantum advice. The idea seems to work even in our case, but we leave it to future work. For details of the idea, see [Kre21].

5 Breaking PRUs

In this section, we show that with probability 1 over $\mathcal{O}$, non-adaptive and $O(\log \lambda)$ -ancilla PRUs do not exist relative to $\mathcal{O}$. For its proof, we construct a QPT adversary that breaks PRUs relative to $\mathcal{O}$.

5.1 Construction of Adversary

Let $G^{\mathcal{O}}$ be a QPT algorithm that satisfies the correctness condition for c -ancilla PRUs. In particular, we consider the case $c(\lambda) = O(\log \lambda)$. For such $G^{\mathcal{O}}$, let $\{U_k\}_{k \in \mathcal{K}_{\lambda}}$ be the unitary implemented by $G^{\mathcal{O}}$ on input $k \in \mathcal{K}_{\lambda}$, where $\mathcal{K}_{\lambda}$ denotes the key-space. We define the following map:

$$\mathcal{M}_{\{U_k\}, \ell}(\cdot) = \mathbb{E}_{k \leftarrow \mathcal{K}_{\lambda}} U_k^{\otimes \ell}(\cdot) U_k^{\dagger \otimes \ell}. \quad (48)$$

Before constructing the adversary, we introduce some lemmas.

Lemma 5.1. *Let $T(\lambda)$ be a polynomial. Let $\epsilon > 0$ and $c, d \in \mathbb{N}$. Let $\{U_k\}_{k \in \mathcal{K}_{\lambda}}$ be an ensemble of λ -qubit unitaries each of which is QPT implementable by making T queries to $\mathcal{O}$ and using c ancilla qubits, where $\mathcal{O}$ is defined in Definition 3.1. For all $n \in [d]$, let $\mathcal{S}'_n$ be any unitary satisfying*

$$\|\mathcal{S}_n(\cdot) \mathcal{S}_n^{\dagger} - \mathcal{S}'_n(\cdot) \mathcal{S}'_n{}^{\dagger}\|_{\diamond} \leq \epsilon. \quad (49)$$

*Then, for any polynomial ℓ , there exists a family $\{V_k\}_{k \in \mathcal{K}_{\lambda}}$ of $(\lambda + c)$ -qubit unitaries such that each V_k is QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and query access to the **UnitaryPSPACE**-complete oracle $\mathcal{U}$ such that it satisfies*

$$\left\| (\mathcal{M}_{\{U_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{E}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O(\ell T \epsilon) + O\left(\frac{2^{c/2} \ell T}{2^{d/2}}\right). \quad (50)$$

Here, $\mathcal{E}_{\{V_k\}, \ell}$ is a CPTP map acting on $\lambda\ell$ qubits defined as follows:

$$\mathcal{E}_{\{V_k\}, \ell}((\cdot)_{\mathbf{A}}) := \text{Tr}_{\mathbf{B}} \left[\mathbb{E}_{k \leftarrow \mathcal{K}_{\lambda}} V_{k, \mathbf{AB}}^{\otimes \ell} ((\cdot)_{\mathbf{A}} \otimes |0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell}) V_{k, \mathbf{AB}}^{\dagger \otimes \ell} \right], \quad (51)$$

where $\mathbf{A}$ is a $\lambda\ell$ -qubit register, and $\mathbf{B}$ is a $c\ell$ -qubit register.

Lemma 5.2. *Let $\{V_k\}_{k \in \mathcal{K}_{\lambda}}$ be a family of $(\lambda + c)$ -qubit unitaries in Lemma 5.1. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_{\lambda}| \rceil$. Then, there exists a QPT algorithm $\mathcal{D}^{\mathcal{U}}$ that, on input classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$, distinguishes $(\mathcal{E}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ from $(\mathcal{M}_{\{U_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ with advantage at least $1 - \text{negl}(\lambda)$.*

Algorithm 1 Adversary distinguishing $\{U_k\}_{k \in \mathcal{K}_\lambda}$ from Haar random unitary.

Oracle access: The algorithm has query access to

- a unitary $W \in \mathbb{U}(2^\lambda)$ which is whether U_k or a Haar random unitary.
- the oracle $\mathcal{O} = (\mathcal{S}, \mathcal{U})$ and its inverse defined in Definition 3.1;

Input: The algorithm takes the security parameter 1^λ as input.

Define $\ell := \lceil \log |\mathcal{K}_\lambda| \rceil$ and $d := 2 \log(\ell T p) + c$, where p is a polynomial, and T is the number of queries to $\mathcal{O}$ to implement U_k .

1. For $n \in [d]$, run the process tomography algorithm in Theorem 2.7 on inputs $\epsilon := \frac{1}{\ell T p}$ and $\eta := 2^{-\lambda-1}$ for $\mathcal{S}_n$ to get a classical description of $\mathcal{S}'_n$. Note that $\|\mathcal{S}_n(\cdot)\mathcal{S}_n^\dagger - \mathcal{S}'_n(\cdot)\mathcal{S}'_n^\dagger\|_\diamond \leq \epsilon$ holds with probability at least $1 - 2^{-\lambda}$ over the randomness of the process tomography algorithm.
2. Prepare $(U^{\otimes \ell} \otimes I) |\Omega_{2^{\lambda \ell}}\rangle$ by querying U .
3. Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be a family of unitaries in Lemma 5.1. Note that each V_k is QPT implementable with access to $\mathcal{U}$ and classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$, where such classical descriptions are obtained in the step 1. Let $\mathcal{D}^{(\cdot)}$ be a QPT algorithm in Lemma 5.2 for $\{V_k\}_{k \in \mathcal{K}_\lambda}$. By querying $\mathcal{U}$, run $\mathcal{D}^{\mathcal{U}}$ on input $(U^{\otimes \ell} \otimes I) |\Omega_{2^{\lambda \ell}}\rangle$ and classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ to get $b \in \{0, 1\}$.

Output: The algorithm outputs b .

With these ingredients at hand, we can now give an adversary to break PRUs, which implies the second item in Theorem 3.3:

Theorem 5.3. Consider $c(\lambda) = O(\log \lambda)$. Let $\mathcal{O}$ be a fixed oracle defined in Definition 3.1. Let $G^\mathcal{O}$ be a QPT algorithm that satisfies the correctness of $c(\lambda)$ -ancilla PRU. For such $G^\mathcal{O}$, let $\{U_k\}_{k \in \mathcal{K}_\lambda}$ be the unitary implemented by $G^\mathcal{O}$ on input $k \in \mathcal{K}_\lambda$, where $\mathcal{K}_\lambda$ denotes the key-space. Then, for any polynomial p , there exists a QPT adversary $\mathcal{A}^{(\cdot, \cdot)}$ such that

$$\left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{U_k, \mathcal{O}}(1^\lambda)] - \Pr_{U \leftarrow \mu_{2^\lambda}} [1 \leftarrow \mathcal{A}^{U, \mathcal{O}}(1^\lambda)] \right| \geq 1 - O\left(\frac{1}{p(\lambda)}\right). \quad (52)$$

Moreover, $\mathcal{A}^{(\cdot, \cdot), \mathcal{O}}$ queries the first oracle non-adaptively.

Proof of Theorem 5.3. We construct $\mathcal{A}^{(\cdot, \cdot)}$ as in Algorithm 1. It is clear that $\mathcal{A}^{(\cdot, \cdot)}$ is a QPT algorithm. Step 1 runs in QPT because $2^d = (\ell T p)^2 + 2^c \leq \text{poly}(\lambda)$, given that $c = O(\log \lambda)$. Steps 2 and 3 also run in QPT, as established in Lemma 5.2.

Assume that the tomography is successful in the step 1, namely, we have $\|\mathcal{S}_n(\cdot)\mathcal{S}_n^\dagger - \mathcal{S}'_n(\cdot)\mathcal{S}'_n^\dagger\|_\diamond \leq \epsilon$ for all $n \in [d]$. For notational simplicity, let E denote this event. Note that

$$\Pr[E] \geq 1 - 2^{-\lambda} \geq 1 - \text{negl}(\lambda) \quad (53)$$

from Theorem 2.7. Thus, it suffices to show that $\mathcal{A}$ can distinguish $\{U_k\}_k$ from Haar random unitaries when the tomography succeeds. Recall that $\{V_k\}_k$ is a family of unitaries in the step 3, and $\mathcal{D}^{(\cdot)}$ is a QPT algorithm

in the step 3 for $\{V_k\}_k$. Note that

$$\Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{U_k, \mathcal{O}}(1^\lambda) | E] = \Pr \left[1 \leftarrow \mathcal{D}^{\mathcal{U}} \left((\mathcal{M}_{\{U_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right) \right] \quad (54)$$

and

$$\Pr_{U \leftarrow \mu_{2\lambda}} [1 \leftarrow \mathcal{A}^{U, \mathcal{O}}(1^\lambda) | E] = \Pr \left[1 \leftarrow \mathcal{D}^{\mathcal{U}} \left((\mathcal{M}_{\mu_{2\lambda}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right) \right]. \quad (55)$$

We prove our claim by the standard hybrid argument. First, we replace $\rho_0 := (\mathcal{M}_{\{U_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ in Equation (54) with $\rho_1 := (\mathcal{E}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$. We obtain the following claim.

Claim 5.4.

$$\left| \Pr [1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho_0)] - \Pr [1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho_1)] \right| \leq O\left(\frac{1}{p(\lambda)}\right).$$

Proof of Claim 5.4. From Lemma 5.1, we have

$$\frac{1}{2} \|\rho_0 - \rho_1\|_1 \leq O(\ell T \epsilon) + O\left(\frac{2^{c/2} \ell T}{2^{d/2}}\right) \leq O\left(\frac{1}{p(\lambda)}\right).$$

Here, $\epsilon = \frac{1}{\ell T p}$ and $d = 2 \log(\ell T p) + c$. □

Next, we replace ρ_1 with $\rho_2 := (\mathcal{M}_{\mu_{2\lambda}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$. We have the following claim. Because its proof is straightforward from Lemma 5.2, we omit it.

Claim 5.5.

$$\left| \Pr [1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho_1)] - \Pr [1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho_2)] \right| \geq 1 - \text{negl}(\lambda).$$

Combing Equations (53) to (55) with Claims 5.4 and 5.5, we have

$$\begin{aligned} & \left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{U_k, \mathcal{O}}(1^\lambda)] - \Pr_{U \leftarrow \mu_{2\lambda}} [1 \leftarrow \mathcal{A}^{U, \mathcal{O}}(1^\lambda)] \right| \\ & \geq \Pr[E] \left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{U_k, \mathcal{O}}(1^\lambda) | E] - \Pr_{U \leftarrow \mu_{2\lambda}} [1 \leftarrow \mathcal{A}^{U, \mathcal{O}}(1^\lambda) | E] \right| - 2 \Pr[\bar{E}] \\ & \geq (1 - \text{negl}(\lambda)) \left(1 - O\left(\frac{1}{p(\lambda)}\right) - \text{negl}(\lambda) \right) - \text{negl}(\lambda) \\ & \geq 1 - O\left(\frac{1}{p(\lambda)}\right), \end{aligned} \quad (56)$$

which concludes the proof. □

5.2 Proof of Lemma 5.1

In this subsection, we prove Lemma 5.1. To show it, we remove queries to $\mathcal{S}_n$ for all small n . To this end, we need the following lemma.

Lemma 5.6. *Let $c' \in \mathbb{N}$. Let $2n + 1 \leq \lambda + c'$. Consider $\mathcal{S}_n$ defined in Definition 3.1. Then, for any $U, V \in \mathbb{U}(2^{\lambda+c'})$,*

$$\frac{1}{2} \|\psi\langle\psi| - \phi\langle\phi|\|_1 \leq O\left(\frac{2^{c'/2}}{2^{n/2}}\right), \quad (57)$$

where

$$|\psi\rangle_{\mathbf{A}\mathbf{A}'\mathbf{B}} := (U(\mathcal{S}_n \otimes I)V)_{\mathbf{A}\mathbf{B}} \otimes I_{\mathbf{A}'} \cdot |\Omega_{2^\lambda}\rangle_{\mathbf{A}\mathbf{A}'} |0^{c'}\rangle_{\mathbf{B}}, \quad (58)$$

$$|\phi\rangle_{\mathbf{A}\mathbf{A}'\mathbf{B}} := (UV)_{\mathbf{A}\mathbf{B}} \otimes I_{\mathbf{A}'} \cdot |\Omega_{2^\lambda}\rangle_{\mathbf{A}\mathbf{A}'} |0^{c'}\rangle_{\mathbf{B}}. \quad (59)$$

Here, $\mathbf{A}, \mathbf{A}'$ are λ -qubit registers, respectively, and $\mathbf{B}$ is a c' -qubit register.

Proof of Lemma 5.6. Define the following states:

$$|\psi'\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'} := (U(\mathcal{S}_n \otimes I)V)_{\mathbf{A}\mathbf{B}} \otimes I_{\mathbf{A}'\mathbf{B}'} \cdot |\Omega_{2^{\lambda+c'}}\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'}, \quad (60)$$

$$|\phi'\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'} := (UV)_{\mathbf{A}\mathbf{B}} \otimes I_{\mathbf{A}'\mathbf{B}'} \cdot |\Omega_{2^{\lambda+c'}}\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'}, \quad (61)$$

where $\mathbf{B}'$ is a c' -qubit register and $|\Omega_{2^{\lambda+c'}}\rangle$ is across $(\mathbf{A}\mathbf{B}, \mathbf{A}'\mathbf{B}')$.

We can prove the following inequality, which we will prove later.

$$\| |\psi'\rangle - |\phi'\rangle \| \leq O\left(\frac{1}{2^{n/2}}\right). \quad (62)$$

Moreover, by definition, it holds that

$$\begin{aligned} |\psi\rangle_{\mathbf{A}\mathbf{A}'\mathbf{B}} &= \sqrt{2^{c'}} \cdot I_{\mathbf{A}\mathbf{A}'\mathbf{B}} \otimes \langle 0^{c'} |_{\mathbf{B}'} \cdot |\psi'\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'}, \\ |\phi\rangle_{\mathbf{A}\mathbf{A}'\mathbf{B}} &= \sqrt{2^{c'}} \cdot I_{\mathbf{A}\mathbf{A}'\mathbf{B}} \otimes \langle 0^{c'} |_{\mathbf{B}'} \cdot |\phi'\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'}. \end{aligned} \quad (63)$$

Putting them together, we obtain

$$\begin{aligned} & \frac{1}{2} \|\psi\langle\psi| - \phi\langle\phi|\|_1 \\ & \leq \| |\psi\rangle - |\phi\rangle \| \quad (\text{Since trace distance between pure states is bounded by their Euclidean distance}) \\ & = \sqrt{2^{c'}} \cdot \| I \otimes \langle 0^{c'} | \cdot (|\psi'\rangle - |\phi'\rangle) \| \quad (\text{By Equation (63)}) \\ & \leq \sqrt{2^{c'}} \cdot \| I \otimes \langle 0^{c'} | \|_\infty \cdot \| |\psi'\rangle - |\phi'\rangle \| \quad (\text{By } \|A|v\rangle\| \leq \|A\|_\infty \|v\rangle\|) \\ & = O\left(\frac{2^{c'/2}}{2^{n/2}}\right) \quad (\text{By } \|I \otimes \langle 0^{c'} | \|_\infty = 1 \text{ and Equation (62)}) \end{aligned}$$

as desired.

To conclude the proof, we prove Equation (62). Define the projection

$$\Pi_n := \sum_{m \in \{0,1\}^n} |m\rangle\langle m| \otimes I_{\perp}^{n,m} \quad (64)$$

onto the subspace on which $\mathcal{S}_n$ acts as the identity. It then follows that

$$(I - \mathcal{S}_n)\Pi_n = 0. \quad (65)$$

First, we can see $(\mathcal{S}_n \otimes I^{\otimes 2\lambda+2c'-2n-1}) |\Omega_{2\lambda+c}\rangle$ is close to $|\Omega_{2\lambda+c}\rangle$ in Euclidean distance as follows:

$$\| |\Omega_{2\lambda+c'}\rangle - (\mathcal{S}_n \otimes I^{\otimes 2\lambda+2c'-2n-1}) |\Omega_{2\lambda+c}\rangle \| \quad (66)$$

$$= \| (I^{\otimes 2n+1} - \mathcal{S}_n) \otimes I^{\otimes 2\lambda+2c'-2n-1} |\Omega_{2\lambda+c'}\rangle \| \quad (67)$$

$$= \| (I^{\otimes 2n+1} - \mathcal{S}_n)(I - \Pi_n) \otimes I^{\otimes 2\lambda+2c'-2n-1} |\Omega_{2\lambda+c'}\rangle \| \quad (\text{By Equation (64)})$$

$$\leq \| (I^{\otimes 2n+1} - \mathcal{S}_n) \otimes I^{\otimes 2\lambda+2c'-2n-1} \|_\infty \cdot \| (I^{\otimes 2n+1} - \Pi_n) \otimes I^{\otimes 2\lambda+2c'-2n-1} |\Omega_{2\lambda+c'}\rangle \|$$

(By $\|A|v\rangle\| \leq \|A\|_\infty \|v\rangle\|$)

$$\leq 2 \| (I^{\otimes 2n+1} - \Pi_n) \otimes I^{\otimes 2\lambda+2c'-2n-1} |\Omega_{2\lambda+c'}\rangle \|$$

(By the triangle inequality and the fact that unitaries have unit operator norm)

$$= 2 \sqrt{\frac{1}{2^{\lambda+c'}} \text{Tr}[(I^{\otimes 2n+1} - \Pi_n) \otimes I^{\otimes 2\lambda+2c'-2n-1}]} \quad (\text{By } \|(A \otimes I)|\Omega_D\rangle\|^2 = \frac{1}{D} \text{Tr}[A^\dagger A] \text{ for any } A \in \mathbb{U}(D))$$

$$= 2 \sqrt{\frac{1}{2^{2n+1}} \text{Tr}[I^{\otimes 2n+1} - \Pi_n]} \quad (\text{By } \text{Tr}[A \otimes B] = \text{Tr}[A]\text{Tr}[B] \text{ for any matrix } A \text{ and } B)$$

$$= 2 \sqrt{\frac{1}{2^{2n+1}} \sum_{m \in \{0,1\}^n} \text{Tr}[|m\rangle\langle m| \otimes (I^{\otimes n+1} - I_\perp^{n,m})]} \quad (\text{By Equation (64)})$$

$$\leq O(2^{n/2}). \quad (68)$$

Here, in the last line, we have used that $\text{Tr}[I^{\otimes n+1} - I_\perp^{n,m}] = 2$. Since $|\psi'\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'} = (U_{\mathbf{A}\mathbf{B}} \otimes V_{\mathbf{A}'\mathbf{B}'}^\top)((\mathcal{S}_n \otimes I)_{\mathbf{A}\mathbf{B}} \otimes I_{\mathbf{A}'\mathbf{B}'}) |\Omega_{2\lambda+c}\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'}$ and $|\phi'\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'} = (U_{\mathbf{A}\mathbf{B}} \otimes V_{\mathbf{A}'\mathbf{B}'}^\top) |\Omega_{2\lambda+c}\rangle_{\mathbf{A}\mathbf{B}\mathbf{A}'\mathbf{B}'}$, we obtain Equation (62), which concludes the proof. $\square$

Having this lemma, we are ready to show Lemma 5.1. For the reader's convenience, we restate it now:

Lemma 5.1. *Let $T(\lambda)$ be a polynomial. Let $\epsilon > 0$ and $c, d \in \mathbb{N}$. Let $\{U_k\}_{k \in \mathcal{K}_\lambda}$ be an ensemble of λ -qubit unitaries each of which is QPT implementable by making T queries to $\mathcal{O}$ and using c ancilla qubits, where $\mathcal{O}$ is defined in Definition 3.1. For all $n \in [d]$, let $\mathcal{S}'_n$ be any unitary satisfying*

$$\|\mathcal{S}_n(\cdot)\mathcal{S}_n^\dagger - \mathcal{S}'_n(\cdot)\mathcal{S}'_n{}^\dagger\|_\diamond \leq \epsilon. \quad (49)$$

Then, for any polynomial ℓ , there exists a family $\{V_k\}_{k \in \mathcal{K}_\lambda}$ of $(\lambda+c)$ -qubit unitaries such that each V_k is QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and query access to the UnitaryPSPACE-complete oracle $\mathcal{U}$ such that it satisfies

$$\left\| (\mathcal{M}_{\{U_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{E}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O(\ell T \epsilon) + O\left(\frac{2^{c/2} \ell T}{2^{d/2}}\right). \quad (50)$$

Here, $\mathcal{E}_{\{V_k\}, \ell}$ is a CPTP map acting on $\lambda\ell$ qubits defined as follows:

$$\mathcal{E}_{\{V_k\}, \ell}((\cdot)_{\mathbf{A}}) := \text{Tr}_{\mathbf{B}} \left[\mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} V_{k, \mathbf{A}\mathbf{B}}^{\otimes \ell} ((\cdot)_{\mathbf{A}} \otimes |0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell}) V_{k, \mathbf{A}\mathbf{B}}^{\dagger \otimes \ell} \right], \quad (51)$$

where $\mathbf{A}$ is a $\lambda\ell$ -qubit register, and $\mathbf{B}$ is a $c\ell$ -qubit register.

Proof of Lemma 5.1. For each $k \in \mathcal{K}_\lambda$, we may view $G^\mathcal{O}(k, \cdot)$ as acting as follows: first, it prepares $|0^c\rangle$ in the ancilla register, then applies a $(\lambda + c)$ -qubit unitary W_k to the input and ancilla qubits by querying $\mathcal{O}$, and finally discards the ancilla register.

We define a $(\lambda + c)$ -qubit unitary V_k as follows. Take the same circuit as in the implementation of W_k , except at the points where it queries $\mathcal{S}_n$. Whenever the circuit queries $\mathcal{S}_n$,

- if $n \in [d]$, apply $\mathcal{S}'_n$ by using its classical description;
- if $n \in [\lambda + c]/[d]$, do not apply any unitary circuit.

The unitary V_k is then the unitary implemented by these modified procedures. Thus, it is clear that V_k is QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and query access to the **UnitaryPSPACE**-complete oracle $\mathcal{U}$.

We prove Equation (50) by the standard hybrid argument. Define a unitary $\tilde{V}_k$ as follows. Take the same circuit as in the implementation of W_k , except at the points where it queries $\mathcal{S}_n$. Whenever the circuit queries $\mathcal{S}_n$,

- if $n \in [d]$, apply $\mathcal{S}_n$;
- if $n \in [\lambda + c]/[d]$, do not apply any unitary.

The unitary $\tilde{V}_k$ is then the unitary implemented by these modified procedures. Define a CPTP map $\mathcal{E}_{\{\tilde{V}_k\}_{k,\ell}}$ acting on $\lambda\ell$ qubits in the same manner as $\mathcal{E}_{\{V_k\}_{k,\ell}}$. We can show

$$\left\| (\mathcal{E}_{\{V_k\}_{k,\ell}} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{E}_{\{\tilde{V}_k\}_{k,\ell}} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O(\ell T \epsilon) \quad (69)$$

and

$$\left\| (\mathcal{E}_{\{\tilde{V}_k\}_{k,\ell}} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{M}_{\{U_k\}_{k,\ell}} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O\left(\frac{2^{c/2}\ell T}{2^{d/2}}\right). \quad (70)$$

We will give the proofs later. From Equations (69) and (70) and the triangle inequality, we obtain Equation (50).

First, we prove Equation (69). For each $k \in \mathcal{K}_\lambda$, the difference between V_k and $\tilde{V}_k$ lies only whether we apply $\mathcal{S}'_n$ or $\mathcal{S}_n$ for all $n \in [d]$. Since the number of queries to the swap unitaries is at most T , we have

$$\|V_k(\cdot)V_k^\dagger - \tilde{V}_k(\cdot)\tilde{V}_k^\dagger\|_\diamond \leq T\epsilon \quad (71)$$

for all $k \in \mathcal{K}_\lambda$, which implies Equation (69).

Next, we prove Equation (70). For each $k \in \mathcal{K}_\lambda$, the difference between W_k and $\tilde{V}_k$ lies only whether we apply $\mathcal{S}_n$ or not for all $n \in [\lambda]/[d]$. Note that the number of queries to the swap unitaries is at most T . Thus, from Lemma 5.6 with $c' = c$, we have

$$\left\| (\tilde{V}_k \otimes I)(|0^c\rangle\langle 0^c| \otimes |\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|)(\tilde{V}_k \otimes I)^\dagger - (W_k \otimes I)(|0^c\rangle\langle 0^c| \otimes |\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|)(W_k \otimes I)^\dagger \right\|_1 \leq O\left(\frac{2^{c/2}T}{2^{d/2}}\right) \quad (72)$$

for all $k \in \mathcal{K}_\lambda$. By combining this inequality and $\|\rho^{\otimes \ell} - \sigma^{\otimes \ell}\|_1 \leq \ell \|\rho - \sigma\|_1$ for any state ρ and σ ²¹, we have

$$\left\| \left(\tilde{V}_{k, \mathbf{BA}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'} \right) (|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{AA}'}) - \left(W_{k, \mathbf{BA}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'} \right) (|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{AA}'}) \right\|_1 \leq O\left(\frac{2^{c/2}\ell T}{2^{d/2}}\right) \quad (73)$$

for all $k \in \mathcal{K}_\lambda$, where $\mathbf{A}$ and $\mathbf{A}'$ are $\lambda\ell$ -qubit register, and $\mathbf{B}$ is a $c\ell$ -qubit register. Here, note that

$$\text{Tr}_{\mathbf{B}}[(W_{k, \mathbf{BA}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'})(|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{AA}'})] = U_{k, \mathbf{A}}^{\otimes \ell}(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{AA}'})U_{k, \mathbf{A}}^{\dagger \otimes \ell} \quad (74)$$

since W_k is a purification of $G^{\mathcal{O}}(k, \cdot)$. Therefore, we obtain Equation (70) from Equation (73). $\square$

5.3 Proof of Lemma 5.2

To show Lemma 5.2, we need some lemmas. First lemma ensures that we can implement the block-encoding of $(\mathcal{E}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)$ by querying $\mathcal{U}$ and by using the classical descriptions of $\mathcal{S}'_n$.

Lemma 5.7. *Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be a family of $(\lambda + c)$ -qubit unitaries that are QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and with the query access to $\mathcal{U}$, where $\mathcal{U}$ is the **UnitaryPSPACE** complete problem in Lemma 2.16. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_\lambda| \rceil$. Then, for any polynomial p , there exists a unitary circuit V_λ satisfying the following:*

- V_λ is QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and with the query access to $\mathcal{U}$.
- V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of $(\mathcal{E}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)$.

Proof of Lemma 5.7. Let x denote the concatenation of classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$. Let A and B be unitaries such that BA is a purification unitary of $(\mathcal{M}_{\{U_k^t\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)$. In other words, tracing out some qubits of $BA|0\dots 0\rangle$ is equal to $(\mathcal{M}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)$. A is a unitary that maps $|0\dots 0\rangle$ to $|x\rangle$. B is the following unitary:

1. First, map $|0\dots 0\rangle |x\rangle \mapsto \frac{1}{\sqrt{|\mathcal{K}_\lambda|}} \sum_{k \in \mathcal{K}_\lambda} |0^c\rangle^{\otimes \ell} |\Omega_{2\lambda\ell}\rangle |0\dots 0\rangle |k\rangle |x\rangle$.
2. Then, map $\frac{1}{\sqrt{|\mathcal{K}_\lambda|}} \sum_{k \in \mathcal{K}_\lambda} |0^c\rangle^{\otimes \ell} |\Omega_{2\lambda\ell}\rangle |0\dots 0\rangle |k\rangle |x\rangle \mapsto \frac{1}{\sqrt{|\mathcal{K}_\lambda|}} \sum_{k \in \mathcal{K}_\lambda} (V_k^{\otimes \ell} \otimes I)(|0^c\rangle^{\otimes \ell} |\Omega_{2\lambda\ell}\rangle) |0\dots 0\rangle |k\rangle |x\rangle$.

Clearly, A is QPT implementable given x . B can be approximately QPT implementable with an exponentially-small error by querying $\mathcal{U}$, because of the following reason: The first step of B is QPT implementable. For the second step of B , we have only to show that each controlled- V_k is approximately QPT implementable by querying $\mathcal{U}$. In fact, first, V_k is QPT implementable on input x, k and by querying $\mathcal{U}$. Second, in order to implement the controlled- V_k , we need the controlled- $\mathcal{U}$. The controlled- $\mathcal{U}$ is in **UnitaryPSPACE** from Remark 2.15, therefore it is approximately QPT implementable with an exponentially-small error by querying $\mathcal{U}$.

Thus, for any polynomial p , there exists a QPT algorithm $\mathcal{B}$ that implements B with error $2^{-p(\lambda)}$ by querying $\mathcal{U}$. Namely, it satisfies

$$\|\mathcal{B}^{\mathcal{U}}(\cdot) - B(\cdot)B^\dagger\|_\diamond \leq 2^{-p(\lambda)}. \quad (75)$$

²¹The latter follows from the triangle inequality.

Since $\mathcal{B}$ is a QPT algorithm, it queries $\mathcal{U}$ at most polynomially many times. Thus, by postponing all intermediate measurements, we can assume that $\mathcal{B}$ applies a QPT unitary C by querying $\mathcal{U}$. Thus, we have

$$\|\text{Tr}_{\mathbf{Y}}[C_{\mathbf{XY}}((\cdot)_{\mathbf{X}} \otimes |0\dots 0\rangle\langle 0\dots 0|_{\mathbf{Y}})C_{\mathbf{XY}}^\dagger] - (B(\cdot)B^\dagger)_{\mathbf{X}}\|_\diamond \leq 2^{-p(\lambda)}, \quad (76)$$

where $\mathbf{X}$ and $\mathbf{Y}$ denote the main register and the ancilla register, respectively. Suppose that $\mathbf{A}$ and $\mathbf{A}'$ are $\lambda\ell$ -qubit registers, and $\mathbf{B}$ is a $c\ell$ -qubit register. We decompose $\mathbf{X}$ as $\mathbf{X}_0 := \mathbf{B}\mathbf{A}\mathbf{A}'$ and $\mathbf{X}_1$, where $\mathbf{X}_0$ is the first $2\lambda\ell$ qubits, and $\mathbf{X}_1$ is the other qubits. From Equation (76) and $\text{Tr}_{\mathbf{X}_1}[(B\mathbf{A}|0\dots 0\rangle\langle 0\dots 0|_{\mathbf{A}^\dagger}B^\dagger)_{\mathbf{X}}] = (\mathcal{M}_{\{V_k\},\ell,\mathbf{B}\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}),$ we have

$$\|\text{Tr}_{\mathbf{B}\mathbf{X}_1\mathbf{Y}}[C_{\mathbf{XY}}A_{\mathbf{X}}(|0\dots 0\rangle\langle 0\dots 0|_{\mathbf{XY}})A_{\mathbf{X}}^\dagger C_{\mathbf{XY}}^\dagger] - (\mathcal{E}_{\{V_k\},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}\|_1 \leq 2^{-p(\lambda)}. \quad (77)$$

Now we are ready to construct block-encoding of $\text{Tr}_{\mathbf{B}}[(\mathcal{M}_{\{V_k\},\ell,\mathbf{B}\mathbf{A}} \otimes \text{id})(|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})]$. From Lemma 2.19, there exists a $(1, 0, \text{poly}(\lambda))$ -block encoding unitary V_λ of

$$\sigma_{\mathbf{A}\mathbf{A}'} := \text{Tr}_{\mathbf{B}\mathbf{X}_1\mathbf{Y}}[C_{\mathbf{XY}}A_{\mathbf{X}}(|0\dots 0\rangle\langle 0\dots 0|_{\mathbf{XY}})A_{\mathbf{X}}^\dagger C_{\mathbf{XY}}^\dagger]. \quad (78)$$

From Lemma 2.19 V_λ can be realized with a single use of $C_{\mathbf{XY}}A_{\mathbf{X}}$ and its inverse, and $\text{poly}(\lambda)$ two-qubit gates. Therefore, V_λ is QPT implementable with x and with the query access to $\mathcal{U}$. Moreover, V_λ satisfies

$$\begin{aligned} & \|((|0^{\text{poly}(\lambda)}\rangle \otimes I)V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I))_{\mathbf{A}\mathbf{A}'} - (\mathcal{E}_{\{V_k\},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}\|_\infty \\ &= \|\sigma_{\mathbf{A}\mathbf{A}'} - (\mathcal{E}_{\{V_k\},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}\|_\infty \quad (\text{Since } V_\lambda \text{ is an } (1, 0, \text{poly}(\lambda))\text{-block encoding of } \sigma) \\ &\leq \|\sigma_{\mathbf{A}\mathbf{A}'} - (\mathcal{E}_{\{V_k\},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}\|_1 \quad (\text{By } \|A\|_\infty \leq \|A\|_1) \\ &\leq 2^{-p(\lambda)}, \quad (\text{By Equation (77)}) \end{aligned}$$

which implies that V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of $(\mathcal{E}_{\{V_k\},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}).$ $\square$

The next lemma ensures that $(\mathcal{M}_{\mu_{2\lambda},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})$ has negligible overlap with the support of $(\mathcal{E}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)$.

Lemma 5.8. *Suppose that $c(\lambda) = O(\log \lambda)$. Let Q be the projection onto the support of $(\mathcal{E}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)$. Then,*

$$\text{Tr}[Q(\mathcal{M}_{\mu_{2\lambda},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|)] \leq \text{negl}(\lambda). \quad (79)$$

Proof of Lemma 5.8. First, we prove that $\text{Tr}[Q] \leq 2^{(1+c)\ell}$. Note that

$$(\mathcal{E}_{\{V_k\},\ell,\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'} = \frac{1}{|\mathcal{K}_\lambda|} \sum_{k \in \mathcal{K}_\lambda} \text{Tr}_{\mathbf{B}}[(V_{k,\mathbf{B}\mathbf{A}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'})(|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}), \quad (80)$$

where $\mathbf{A}$ and $\mathbf{A}'$ are $\lambda\ell$ -qubit registers, and $\mathbf{B}$ is a $c\ell$ -qubit register. For each k , the rank of $\text{Tr}_{\mathbf{B}}[(V_{k,\mathbf{B}\mathbf{A}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'})(|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'}]$ is at most $\min\{2^{2\ell}, 2^{2\lambda\ell}\} = 2^{c\ell}$ since $(V_{k,\mathbf{B}\mathbf{A}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'})(|0^c\rangle\langle 0^c|_{\mathbf{B}}^{\otimes \ell} \otimes |\Omega_{2\lambda\ell}\rangle\langle \Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})$ is pure. Thus, the rank of Q is at most $2^{c\ell} \cdot |\mathcal{K}_\lambda| \leq 2^{(1+c)\ell}$, which implies $\text{Tr}[Q] \leq 2^{(1+c)\ell}$.

Having this,

$$\begin{aligned}
\text{Tr}[Q(\mathcal{M}_{\mu_{2\lambda},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)] &\leq \text{Tr}[Q \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{2\lambda}} |\psi\rangle\langle\psi|^{\otimes \ell}] + O\left(\frac{\ell^2}{2^\lambda}\right) && \text{(By Lemma 2.12)} \\
&= \frac{\text{Tr}[Q\Pi_{\text{sym}}]}{\binom{2^{2\lambda}+\ell-1}{\ell}} + \text{negl}(\lambda) \\
&\leq \frac{2^{(1+c)\ell}}{\binom{2^{2\lambda}+\ell-1}{\ell}} + \text{negl}(\lambda) && \text{(By } \text{Tr}[Q\Pi_{\text{sym}}] \leq \text{Tr}[Q] \leq 2^{(1+c)\ell}) \\
&\leq O\left(\frac{2^{(1+c)\ell}(\ell!)}{2^{2\lambda\ell}}\right) + \text{negl}(\lambda) \\
&\leq O\left(\frac{2^{(1+c)\ell}\ell^{\ell+1/2}e^{-\ell}}{2^{2\lambda\ell}}\right) + \text{negl}(\lambda) && \text{(By the Stirling's formula, } \ell! \leq \ell^{\ell+1/2}e^{-\ell+1}) \\
&= O\left(\ell^{1/2}\left(\frac{2^{1+c}e^{-1}\ell}{2^{2\lambda}}\right)^\ell\right) + \text{negl}(\lambda) \\
&= O\left(\ell^{1/2}\left(\frac{\text{poly}(\lambda)}{2^{2\lambda}}\right)^\ell\right) + \text{negl}(\lambda) && \text{(By } c(\lambda) = O(\log \lambda)) \\
&\leq \text{negl}(\lambda), && (81)
\end{aligned}$$

which concludes the proof. $\square$

The following lemma gives us an algorithm that distinguishes between two states if they are statistically far.

Lemma 5.9. *Suppose that ρ and ξ are $n(\lambda) = \text{poly}(\lambda)$ -qubit states, and satisfy the following:*

- *For any polynomial p , there exists a QPT implementable unitary V_λ with classical advice c and with query access to $\mathcal{U}$ such that V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ .*
- *For the projection Q onto the support of ρ , $\text{Tr}[Q\xi] \leq \text{negl}(\lambda)$.*

Then, there exists a QPT algorithm $\mathcal{D}^{\mathcal{U}}$ that, on input c , distinguishes ρ from ξ with advantage at least $1 - \text{negl}(\lambda)$.

Before giving the proof, we show Lemma 5.2 with Lemma 5.9. We restate it here for the reader's convenience.

Lemma 5.2. *Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be a family of $(\lambda + c)$ -qubit unitaries in Lemma 5.1. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_\lambda| \rceil$. Then, there exists a QPT algorithm $\mathcal{D}^{\mathcal{U}}$ that, on input classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$, distinguishes $(\mathcal{E}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ from $(\mathcal{M}_{\{U_k\},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ with advantage at least $1 - \text{negl}(\lambda)$.*

Proof of Lemma 5.2. Let $\rho = (\mathcal{E}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ and $\xi = (\mathcal{M}_{\mu_{2\lambda},\ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$. From Lemmata 5.7 and 5.8, they satisfy the condition in Lemma 5.9. Therefore, we obtain Lemma 5.2 by applying Lemma 5.9, which concludes the proof. $\square$

Finally, we give the proof of Lemma 5.9.

Proof of Lemma 5.9. Suppose that ρ and σ are n -qubit state, where $n = \text{poly}(\lambda)$. Let V_λ be a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ , where we chose a polynomial p later. We construct $\mathcal{D}^\mathcal{U}$ from this V_λ . Before that, consider the singular value discrimination algorithm in Theorem 2.17 with $M = (|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)V_\lambda(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)$, $\eta = 2^{-\lambda}$, $a = 2^{-3n}$ and $b = 2^{-2n}$ by querying $\mathcal{U}$. Based on this, we construct $\mathcal{D}^\mathcal{U}$ as follows:

1. Take n -qubit state σ as an input.
2. Simulate the above singular value discrimination algorithm on input $|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes \sigma$. (Because this simulation queries $\mathcal{U}$, it causes at most a negligible error.)

Our goal is to show $\mathcal{D}^\mathcal{U}$ distinguishes ρ from ξ . To use the singular value discrimination algorithm (Theorem 2.17), the input state has to be inside of the promise with high probability. In other words, the input state must have a sufficiently large overlap with the subspace W_0 or W_1 . Here, W_0 is the subspace spanned by the all right singular value vectors of $(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)V_\lambda(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)$ with singular value is at most $a = 2^{-3n}$, W_1 is the subspace spanned by the all right singular value vectors of $(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)V_\lambda(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)$ with singular value is at least $b = 2^{-2n}$. In the following, we show that ρ has at least $1 - \text{negl}(\lambda)$ overlap with W_1 and that ξ has at least $1 - \text{negl}(\lambda)$ overlap with W_0 .

Large overlap with W_1 . First, we show that ρ has a large overlap with W_1 . This follows from that V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ . The formal statement is the following.

Lemma 5.10. *Let $\Pi_{\geq \epsilon}$ be the projection onto the subspace spanned by right singular vectors of $(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)V_\lambda(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)$ with singular value at least ϵ . If V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ , then*

$$\text{Tr}[\Pi_{\geq \epsilon} \rho] \geq 1 - 2^{n-p+1} - 2^n \epsilon. \quad (82)$$

We give its proof later. We define

$$\rho' := \frac{\Pi_{\geq 2^{-2n}} \rho \Pi_{\geq 2^{-2n}}}{\text{Tr}[\Pi_{\geq 2^{-2n}} \rho]}, \quad (83)$$

where $\Pi_{\geq 2^{-2n}}$ is the projection in Lemma 5.10 with $\epsilon = 2^{-2n}$. Therefore, we have

$$\begin{aligned} & \left| \Pr[1 \leftarrow \mathcal{D}^\mathcal{U}(\rho)] - \Pr[1 \leftarrow \mathcal{D}^\mathcal{U}(\rho')] \right| \\ & \leq \|\rho - \rho'\|_1 \\ & \leq \sqrt{2^{n-p+1} + 2^{-n}} \quad (\text{By Lemma 5.10 and the gentle measurement lemma (Lemma 2.1)}) \\ & \leq \sqrt{2^{n-p+1} + \text{negl}(\lambda)}. \end{aligned} \quad (84)$$

Large overlap with W_0 . Next, we show ξ has a large overlap with W_0 . From the assumption, we have

$$\text{Tr}[Q\xi] \leq \text{negl}(\lambda), \quad (85)$$

where Q is the projection onto the support of ρ . We define

$$\xi' := \frac{(I - Q)\xi(I - Q)}{\text{Tr}[(I - Q)\xi]}. \quad (86)$$

With Equation (85) and the gentle measurement lemma (Lemma 2.1), we have

$$\left| \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi)] - \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi')] \right| \leq \|\xi - \xi'\|_1 = \text{negl}(\lambda). \quad (87)$$

Moreover, we can show that ξ' has a large overlap with W_0 from the following lemma. We give its proof later.

Lemma 5.11. *Let $|\psi\rangle$ be a state such that $Q|\psi\rangle = 0$, where Q is the projection onto the support of ρ . Let $\Pi_{\geq \epsilon}$ be the projection in Lemma 5.10. If V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ , then, $\|\Pi_{\geq \epsilon}|\psi\rangle\| \leq 2^{-p}\epsilon^{-1}$.*

Since $\text{Tr}[Q\xi'] = 0$ by its definition (Equation (86)), by applying Lemma 5.11 with $\epsilon = 2^{-3n}$ we have

$$\text{Tr}[\Pi_{\geq 2^{-3n}}\xi'] \leq 2^{-2p+6n}. \quad (88)$$

Thus, let us define

$$\xi'' := \frac{(I - \Pi_{\geq 2^{-3n}})\xi'(I - \Pi_{\geq 2^{-3n}})}{\text{Tr}[(I - \Pi_{\geq 2^{-3n}})\tau]}. \quad (89)$$

From the above inequality and the gentle measurement lemma (Lemma 2.1), we have

$$\left| \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi')] - \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi'')] \right| \leq \|\xi' - \xi''\|_1 \leq 2^{-p+3n}. \quad (90)$$

Combining All Components. Now we are ready to show $\mathcal{D}^{\mathcal{U}}$ distinguishes ρ from ξ . Recall that $\mathcal{D}^{\mathcal{U}}$ simulates the singular value discrimination algorithm in Theorem 2.17 with $M = (|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)V_\lambda(|0^{\text{poly}(\lambda)}\rangle\langle 0^{\text{poly}(\lambda)}| \otimes I)$, $\eta = 2^{-\lambda}$, $a = 2^{-3n}$ and $b = 2^{-2n}$ by querying $\mathcal{U}$. Note that ρ' is on W_1 and ξ'' is on W_0 . Thus, from Theorem 2.17, we have

$$\left| \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho')] - \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi'')] \right| \geq 1 - \text{negl}(\lambda). \quad (91)$$

Moreover, by choosing $p(\lambda) = 4n(\lambda)$, we have

$$\begin{aligned} \left| \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho)] - \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho')] \right| &\leq \sqrt{2^{-3n+1} + \text{negl}(\lambda)} && \text{(By Equation (84))} \\ &\leq \text{negl}(\lambda) && (92) \end{aligned}$$

and

$$\begin{aligned} \left| \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi)] - \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi'')] \right| &\leq \text{negl}(\lambda) + 2^{-n} && \text{(By Equations (87) and (90))} \\ &\leq \text{negl}(\lambda). && (93) \end{aligned}$$

With these inequalities at hand, we have

$$\begin{aligned} &\left| \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\rho)] - \Pr[1 \leftarrow \mathcal{D}^{\mathcal{U}}(\xi)] \right| \\ &\geq 1 - \text{negl}(\lambda), \end{aligned} \quad (94)$$

which concludes the proof. $\square$

We give proofs of Lemmata 5.10 and 5.11 to complete the proof. First, we show Lemma 5.10. We restate it here for the reader's convenience.

Lemma 5.10. *Let $\Pi_{\geq \epsilon}$ be the projection onto the subspace spanned by right singular vectors of $(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I)$ with singular value at least ϵ . If V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ , then*

$$\text{Tr}[\Pi_{\geq \epsilon} \rho] \geq 1 - 2^{n-p+1} - 2^n \epsilon. \quad (82)$$

Proof of Lemma 5.10. We have

$$\text{Tr}[\rho \Pi_{\geq \epsilon}] \quad (95)$$

$$= \left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \Pi_{\geq \epsilon}] + \text{Tr}[(\rho - (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I)) \Pi_{\geq \epsilon}] \right| \quad (96)$$

$$\geq \left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \Pi_{\geq \epsilon}] \right| - \left\| \rho - (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \Pi_{\geq \epsilon} \right\|_1$$

(By the triangle inequality and $|\text{Tr}[A]| \leq \|A\|_1$)

$$\geq \left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \Pi_{\geq \epsilon}] \right| - \left\| \rho - (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \right\|_1.$$

(By Hölder's inequality (Lemma 2.2) and $\|\Pi_{\geq \epsilon}\|_\infty = 1$)

We can show

$$\left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \Pi_{\geq \epsilon}] \right| \geq 1 - 2^{n-p} - 2^n \epsilon \quad (97)$$

and

$$\left\| \rho - (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \right\|_1 \leq 2^{n-p}. \quad (98)$$

We give their proofs later. With these inequalities at hand, we obtain Lemma 5.10.

To conclude the proof, we give proofs of Equations (97) and (98). We can show the latter as follows:

$$\begin{aligned} & \left\| \rho - (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \right\|_1 \\ & \leq 2^n \left\| \rho - (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \right\|_\infty \quad (\text{By } \|A\|_1 \leq d \|A\|_\infty \text{ for any } A \in \mathcal{L}(d)) \\ & \leq 2^{n-p}. \quad (\text{Since } V_\lambda \text{ is a } (1, 2^{-p}, \text{poly}(\lambda))\text{-block encoding of } \rho) \end{aligned} \quad (99)$$

We can show the former as follows.

$$\left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) \Pi_{\geq \epsilon}] \right| \quad (100)$$

$$= \left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I)] - \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) (I - \Pi_{\geq \epsilon})] \right| \quad (101)$$

$$\geq \left| \text{Tr}[(\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I)] - \left\| (\langle 0^{\text{poly}(\lambda)} | \otimes I) V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) (I - \Pi_{\geq \epsilon}) \right\|_1 \right|, \quad (102)$$

where we have used the triangle inequality and $|\text{Tr}[A]| \leq \|A\|_1 = \| -A \|_1$ in the inequality. The first term can be estimated as follows:

$$\begin{aligned}
& \left| \text{Tr}[\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)] \right| \\
&= \left| \text{Tr}[\rho] - \text{Tr}[\rho - \langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)] \right| \\
&\geq |\text{Tr}[\rho]| - \left\| \rho - \langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I) \right\|_1 \quad (\text{By the triangle inequality and } |\text{Tr}[A]| \leq \|A\|_1) \\
&\geq 1 - 2^{-p}, \tag{103}
\end{aligned}$$

where we have used Equation (98) in the last inequality. To estimate the second term in Equation (102), recall that $\Pi_{\geq \epsilon}$ is the projection onto the subspace spanned by right singular vectors of $\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)$ whose singular values are at least ϵ . Thus, $I - \Pi_{\geq \epsilon}$ is the projection onto the subspace spanned by right singular vectors of $\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)$ whose singular values are less than ϵ . In addition to that, note that the number of singular values of $\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)$ is at most 2^n since $\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)$ is an operator acting on n qubits. With these observations, we have

$$\left\| (\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)(I - \Pi_{\geq \epsilon}) \right\|_1 \leq 2^n \epsilon \tag{104}$$

since $\|A\|_1$ is equivalent to the sum of its singular values. From Equations (102) to (104), we obtain Equation (97). $\square$

Finally, we show Lemma 5.11. We also restate it here.

Lemma 5.11. *Let $|\psi\rangle$ be a state such that $Q|\psi\rangle = 0$, where Q is the projection onto the support of ρ . Let $\Pi_{\geq \epsilon}$ be the projection in Lemma 5.10. If V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of ρ , then, $\|\Pi_{\geq \epsilon}|\psi\rangle\| \leq 2^{-p}\epsilon^{-1}$.*

Proof of Lemma 5.11. Since Q is the projection onto ρ , we have $\rho|\psi\rangle = \rho Q|\psi\rangle = 0$. Then, we have

$$\| \langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I) |\psi\rangle \| = \| (\rho - \langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)) |\psi\rangle \| \tag{105}$$

$$\leq \| \rho - \langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I) \|_\infty \tag{106}$$

$$\leq 2^{-p}, \tag{107}$$

where the last inequality follows from the assumption that V_λ is a $(1, 2^{-p}, \text{poly}(\lambda))$ -block encoding of ρ . Let

$$(\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I) = \sum_i a_i |w_i\rangle \langle v_i| \tag{108}$$

be the singular value decomposition. Namely, $\{|w_i\rangle\}_i$ and $\{|v_i\rangle\}_i$ are sets of orthonormal states, and all a_i are positive real numbers. Since each $|v_i\rangle$ is the right singular vector of $\langle 0^{\text{poly}(\lambda)} | \otimes I V_\lambda(|0^{\text{poly}(\lambda)}\rangle \otimes I)$

whose singular value is a_i , we have $\Pi_{\geq \epsilon} = \sum_{i:a_i \geq \epsilon} |v_i\rangle\langle v_i|$. Then, we have

$$\| \langle 0^{\text{poly}(\lambda)} | \otimes I \rangle V_\lambda (| 0^{\text{poly}(\lambda)} \rangle \otimes I) | \psi \rangle \| = \left\| \sum_i a_i |w_i\rangle \langle v_i | \psi \rangle \right\| \quad (109)$$

$$= \sqrt{\sum_i a_i^2 | \langle v_i | \psi \rangle |^2} \quad (110)$$

$$\geq \sqrt{\sum_{i:a_i \geq \epsilon} a_i^2 | \langle v_i | \psi \rangle |^2} \quad (111)$$

$$\geq \epsilon \sqrt{\sum_{i:a_i \geq \epsilon} | \langle v_i | \psi \rangle |^2} \quad (112)$$

$$= \epsilon \| \Pi_{\geq \epsilon} | \psi \rangle \|, \quad (113)$$

where in the last inequality we have used $\Pi_{\geq \epsilon} = \sum_{i:a_i \geq \epsilon} |v_i\rangle\langle v_i|$. From Equations (107) and (113), we have $\epsilon \| \Pi_{\geq \epsilon} | \psi \rangle \| \leq 2^{-p}$, which implies $\| \Pi_{\geq \epsilon} | \psi \rangle \| \leq 2^{-p} \epsilon^{-1}$. $\square$

6 Oracle Separation Between PRIs with Short Stretch and PRFSGs

In this section, we prove the following.

Theorem 6.1. *Then, with probability 1 over the choice of $\mathcal{O}$ defined in Definition 3.1, the following are satisfied:*

- *Quantumly-accessible adaptively-secure PRFSGs exist relative to $\mathcal{O}$.*
- *Non-adaptive, $O(\log \lambda)$ -ancilla PRIs with $O(\log \lambda)$ stretch do not exist relative to $\mathcal{O}$.*

Remark 6.2. Theorem 6.1 is stronger than Theorem 3.3 because, for any s and c , non-adaptive, c -ancilla PRIs with s stretch are constructed from non-adaptive, c -ancilla PRUs in a black-box manner.

Since we have already proved the first item in Theorem 4.2, it suffices to prove the second item by constructing an adversary breaking PRIs. We give the adversary in a similar way as in Algorithm 1. Let $G^\mathcal{O}$ be a QPT algorithm that satisfies the correctness of c -ancilla PRIs with s stretch. In particular, we consider the case when $c(\lambda) = O(\log \lambda)$ and $s(\lambda) = O(\log \lambda)$. For such $G^\mathcal{O}$, let $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ be the isometry implemented by $G^\mathcal{O}$ on input $k \in \mathcal{K}_\lambda$, where $\mathcal{K}_\lambda$ denotes the key-space. We define the following map:

$$\mathcal{M}_{\{\mathcal{I}_k\}, \ell}(\cdot) = \mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} \mathcal{I}_k^{\otimes \ell}(\cdot) \mathcal{I}_k^{\dagger \otimes \ell}. \quad (114)$$

To construct a PRI adversary, we need two lemmas as in Section 5. These lemmas can be obtained by modifying Lemma 5.1 and Lemma 5.9 for PRIs. We give their proofs later.

Lemma 6.3. *Let $T(\lambda)$ be a polynomial. Let $\epsilon > 0$ and $c, d, s \in \mathbb{N}$. Let $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ be an ensemble of isometries mapping λ qubits to $\lambda + s$ qubits, where each $\mathcal{I}_k$ is QPT implementable with $s + c$ ancilla qubits by making T queries to $\mathcal{O}$ defined in Definition 3.1. For all $n \in [d]$, let $\mathcal{S}'_n$ be any unitary satisfying*

$$\| \mathcal{S}_n(\cdot) \mathcal{S}_n^\dagger - \mathcal{S}'_n(\cdot) \mathcal{S}'_n{}^\dagger \|_\diamond \leq \epsilon. \quad (115)$$

Then, for any polynomial ℓ , there exists a family $\{V_k\}_{k \in \mathcal{K}_\lambda}$ of $(\lambda + s + c)$ -qubit unitaries such that each V_k is QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and query access to the UnitaryPSPACE-complete oracle $\mathcal{U}$ such that it satisfies

$$\left\| (\mathcal{M}_{\{\mathcal{I}_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{F}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O(\ell T \epsilon) + O\left(\frac{2^{s+3c/2}\ell T}{2^{d/2}}\right). \quad (116)$$

Here, $\mathcal{F}_{\{V_k\}, \ell}$ is a CPTP map from $\lambda\ell$ qubits to $(\lambda + s)\ell$ qubits defined as follows:

$$\mathcal{F}_{\{V_k\}, \ell}((\cdot)_{\mathbf{A}}) := \text{Tr}_{\mathbf{C}} \left[\mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} V_{k, \mathbf{ABC}}^{\otimes \ell} ((\cdot)_{\mathbf{A}} \otimes |0^s\rangle\langle 0^s|_{\mathbf{B}}^{\otimes \ell} \otimes |0^c\rangle\langle 0^c|_{\mathbf{C}}^{\otimes \ell}) V_{k, \mathbf{ABC}}^{\dagger \otimes \ell} \right], \quad (117)$$

where $\mathbf{A}$ is a $\lambda\ell$ -qubit register, $\mathbf{B}$ is a $s\ell$ -qubit register, and $\mathbf{C}$ is a $c\ell$ -qubit register.

For the next lemma, we define the Haar random isometry map as follows:

Definition 6.4 (Haar Random Isometry Map). We define²²

$$\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell}(\rho_{\mathbf{A}}) := \mathbb{E}_{U \leftarrow \mu_{2\lambda+s}} \left(\bigotimes_{i \in [\ell]} U_{\mathbf{A}_i \mathbf{B}_i} \right) (\rho_{\mathbf{A}} \otimes |0^s\rangle\langle 0^s|_{\mathbf{B}}^{\otimes \ell}) \left(\bigotimes_{i \in [\ell]} U_{\mathbf{A}_i \mathbf{B}_i} \right)^\dagger, \quad (118)$$

where $\mathbf{A} := \bigotimes_{i \in [\ell]} \mathbf{A}_i$ and $\mathbf{B} := \bigotimes_{i \in [\ell]} \mathbf{B}_i$, where, for each $i \in [\ell]$, $\mathbf{A}_i$ and $\mathbf{B}_i$ are λ -qubit register and s -qubit register, respectively.

Now we are ready to give the following lemma.

Lemma 6.5. Suppose that $c(\lambda) = O(\log \lambda)$. Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be the family of $(\lambda + s + c)$ -qubit unitary, and $\mathcal{F}_{\{V_k\}, \ell}$ be the CPTP map in Lemma 6.3. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_\lambda| \rceil$. Then, there exists a QPT algorithm $\mathcal{D}^{\mathcal{U}}$ that, on input classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$, distinguishes $(\mathcal{F}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ from $(\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ with advantage at least $1 - \text{negl}(\lambda)$.

Based on these lemmas, we construct a PRI adversary as shown in Algorithm 2. The red-highlighted lines in Algorithm 2 show the differences from Algorithm 1.

The following Theorem 6.6 implies Theorem 6.1. We omit the proof of Theorem 6.6 as it is identical to that of Theorem 5.3, except that Lemmata 5.1 and 5.9 are respectively replaced by Lemmata 6.3 and 6.5.

Theorem 6.6. Suppose that $c(\lambda) = O(\log \lambda)$ and $s(\lambda) = O(\log \lambda)$. Let $\mathcal{O}$ be a fixed oracle defined in Definition 3.1. Let $G^{\mathcal{O}}$ be a QPT algorithm that satisfies the correctness of c -ancilla PRIs with s stretch. For such $G^{\mathcal{O}}$, let $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ be the isometry mapping λ qubits to $\lambda + s$ qubits implemented by $G^{\mathcal{O}}$ on input $k \in \mathcal{K}_\lambda$, where $\mathcal{K}_\lambda$ denotes the key-space. Then, for any polynomial p , the QPT adversary $\mathcal{A}^{(\cdot, \cdot)}$ defined in Algorithm 2 satisfies

$$\left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{I}_k, \mathcal{O}}(1^\lambda)] - \Pr_{U \leftarrow \mu_{2\lambda+s}} [1 \leftarrow \mathcal{A}^{\mathcal{I}_U, \mathcal{O}}(1^\lambda)] \right| \geq 1 - O\left(\frac{1}{p(\lambda)}\right), \quad (119)$$

where, for each $U \in \mathbb{U}(2^{\lambda+s(\lambda)})$, $\mathcal{I}_U$ is the isometry that maps λ -qubit state $|\psi\rangle$ to $(\lambda + s(\lambda))$ -qubit state $U(|\psi\rangle|0^s\rangle)$. Here, $\mathcal{A}^{(\cdot, \cdot), \mathcal{O}}$ queries not only $\mathcal{O}$ but also its inverse. Moreover, $\mathcal{A}^{(\cdot, \cdot), \mathcal{O}}$ queries the first oracle non-adaptively.

Remark 6.7. The parameter d is chosen so that $\ell T 2^{s+3c/2-d/2} = O(1/p)$, representing the error term in Lemma 6.3. Algorithm 2 is QPT because $2^d = (\ell T p)^2 \cdot 2^{2s+3c} \leq \text{poly}(\lambda)$, where the inequality holds under the assumption that $c(\lambda) = O(\log \lambda)$ and $s(\lambda) = O(\log \lambda)$. In contrast, if $c(\lambda) = \omega(\log \lambda)$ or $s(\lambda) = \omega(\log \lambda)$, then Algorithm 2 would no longer be QPT.

²²Here, we chose $|0^s\rangle$ as an input state. This does not lose any generality due to the right invariance of the Haar measure.

Algorithm 2 Adversary distinguishing $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ from Haar random isometry relative to $(\mathcal{S}, \mathcal{U})$.

Oracle access: The algorithm has query access to

- an isometry $\mathcal{I}$ mapping λ qubits to $\lambda + s(\lambda)$ qubits, which is whether $\mathcal{I}_k$ or a Haar random isometry.
- the oracle $\mathcal{O} = (\mathcal{S}, \mathcal{U})$ and its inverse defined in Definition 3.1;

Input: The algorithm takes the security parameter 1^λ as input.

Define $\ell := \lceil \log |\mathcal{K}_\lambda| \rceil$ and $d := 2 \log(\ell T p) + 3c + 2s$, where p is a polynomial, and T is the number of queries to $\mathcal{O}$ to implement $\mathcal{I}_k$.

1. For $n \in [d]$, run the process tomography algorithm in Theorem 2.7 on inputs $\epsilon := \frac{1}{\ell T p}$ and $\eta := 2^{-\lambda-1}$ for $\mathcal{S}_n$ to get a classical description of $\mathcal{S}'_n$. Note that $\|\mathcal{S}_n(\cdot)\mathcal{S}_n^\dagger - \mathcal{S}'_n(\cdot)\mathcal{S}'_n{}^\dagger\|_\diamond \leq \epsilon$ holds with probability at least $1 - 2^{-\lambda}$ over the randomness of the process tomography algorithm.
2. Prepare $(\mathcal{I}^{\otimes \ell} \otimes I) |\Omega_{2^{\lambda \ell}}\rangle$ by querying $\mathcal{I}$.
3. Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be a family of unitaries in Lemma 6.3. Note that each V_k is QPT implementable with access to $\mathcal{U}$ and classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$, where such classical descriptions are obtained in the step 1. Let $\mathcal{D}^{(\cdot)}$ be a QPT algorithm in Lemma 6.5 for $\{V_k\}_{k \in \mathcal{K}_\lambda}$. By querying $\mathcal{U}$, run $\mathcal{D}^\mathcal{U}$ on input $(\mathcal{I}^{\otimes \ell} \otimes I) |\Omega_{2^{\lambda \ell}}\rangle$ and classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ to get $b \in \{0, 1\}$.

Output: The algorithm outputs b .

6.1 Proof of Lemma 6.3

In this subsection, we prove Lemma 6.3. The proof strategy is the same as that of Lemma 5.1.

Proof of Lemma 6.3. Let $G^\mathcal{O}(k, \cdot)$ be an algorithm implementing $\mathcal{I}_k$. For each $k \in \mathcal{K}_\lambda$, we may view $G^\mathcal{O}(k, \cdot)$ as acting as follows: first, it prepares $|0^s\rangle |0^c\rangle$ in the ancilla register, then applies a $(\lambda + s + c)$ -qubit unitary W_k to the input and ancilla qubits by querying $\mathcal{O}$, and finally discards the last c qubits in the ancilla register.

We define a $(\lambda + s + c)$ -qubit unitary V_k as follows. Take the same circuit as in the implementation of W_k , except at the points where it queries $\mathcal{S}_n$. Whenever the circuit queries $\mathcal{S}_n$,

- if $n \in [d]$, apply $\mathcal{S}'_n$ by using its classical description;
- if $n \in [\lambda + s + c]/[d]$, do not apply any unitary circuit.

We prove Equation (116) by the standard hybrid argument. For each $k \in \mathcal{K}_\lambda$, we define a $(\lambda + s + c)$ -qubit unitary $\tilde{V}_k$ as follows: apply the same unitary as W_k except for querying $\mathcal{S}_n$. When querying to $\mathcal{S}_n$,

- if $n \in [d]$, apply $\mathcal{S}_n$;
- if $n \in [\lambda + s + c]/[d]$, do not apply any unitary.

Define a CPTP map $\mathcal{F}_{\{\tilde{V}_k\}_{k, \ell}}$ from $\lambda \ell$ qubits to $(\lambda + s)\ell$ qubits in the same manner as $\mathcal{F}_{\{V_k\}_{k, \ell}}$. We can show

$$\left\| (\mathcal{F}_{\{V_k\}_{k, \ell}} \otimes \text{id})(|\Omega_{2^{\lambda \ell}}\rangle\langle\Omega_{2^{\lambda \ell}}|) - (\mathcal{F}_{\{\tilde{V}_k\}_{k, \ell}} \otimes \text{id})(|\Omega_{2^{\lambda \ell}}\rangle\langle\Omega_{2^{\lambda \ell}}|) \right\|_1 \leq O(\ell T \epsilon) \quad (120)$$

and

$$\left\| (\mathcal{F}_{\{\tilde{V}_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{M}_{\{\mathcal{I}_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O(\ell T 2^{s+3c/2-d/2}). \quad (121)$$

We will give the proofs later. From Equations (120) and (121) and the triangle inequality, we obtain Equation (116).

First, we prove Equation (120). For each $k \in \mathcal{K}_\lambda$, the difference between V_k and $\tilde{V}_k$ lies only whether we apply $\mathcal{S}'_n$ or $\mathcal{S}_n$ for all $n \in [d]$. Since the number of queries to the swap unitaries is at most T , we have

$$\|V_k(\cdot)V_k^\dagger - \tilde{V}_k(\cdot)\tilde{V}_k^\dagger\|_\diamond \leq T\epsilon \quad (122)$$

for all $k \in \mathcal{K}_\lambda$, which implies Equation (120).

For each k , define a map $\mathcal{F}_k$ as follows:

$$\mathcal{F}_k(\rho_{\mathbf{X}}) := \text{Tr}_{\mathbf{Z}}[V_{k, \mathbf{XYZ}}(\rho_{\mathbf{X}} \otimes |0^s\rangle\langle 0^s|_{\mathbf{Y}} \otimes |0^c\rangle\langle 0^c|_{\mathbf{Z}})V_{k, \mathbf{XYZ}}^\dagger], \quad (123)$$

where $\mathbf{X}$ is a λ -qubit register, $\mathbf{Y}$ is a s -qubit register, and $\mathbf{Z}$ is a c -qubit register. Suppose that we have

$$\left\| (\mathcal{F}_{k, \mathbf{X} \rightarrow \mathbf{XY}} \otimes \text{id}_{\mathbf{X}'})(|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|_{\mathbf{XX}'} - (\mathcal{I}_{k, \mathbf{X} \rightarrow \mathbf{XY}} \otimes \text{id}_{\mathbf{X}'})(|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|_{\mathbf{XX}'} \right\|_1 \leq O(T 2^{s+3c/2-d/2}) \quad (124)$$

for all $k \in \mathcal{K}_\lambda$, where $\mathbf{X}'$ is a λ -qubit register. Since Equation (124) implies Equation (121), it suffices to prove Equation (124). From the c -ancilla correctness condition, we can view that $\mathcal{I}_k$ is implemented as follows:²³

1. Prepare $|0^s\rangle_{\mathbf{Y}} |0^c\rangle_{\mathbf{Z}}$ on ancilla qubits, and apply $(\lambda + s + c)$ -qubit unitary on $\mathbf{XYZ}$. This is equal to applying an isometry A mapping $\mathbf{X}$ to $\mathbf{XYZ}$.
2. For each $i \in [T]$, perform the following: Apply $\mathcal{T}_i := \mathcal{S}_{n_i}$ onto some $2n_i + 1$ qubits of $\mathbf{XYZ}$, where $2n_i + 1 \leq \lambda + s + c$. Then, apply a $(\lambda + s + c)$ -qubit unitary B_i on $\mathbf{XYZ}$.
3. Discard $\mathbf{Z}$.

From the above observation, we define the following hybrids for each $i \in [T]$:²⁴

$$|\psi_i\rangle_{\mathbf{XYZX}'} = \left(\prod_{j=i+1}^T (B_j \mathcal{T}'_j)_{\mathbf{XYZ}} \prod_{j=1}^i (B_j \mathcal{T}_j)_{\mathbf{XYZ}} A_{\mathbf{X} \rightarrow \mathbf{XYZ}} \otimes I_{\mathbf{X}'} \right) |\Omega_{2\lambda}\rangle_{\mathbf{XX}'}, \quad (125)$$

where $\mathcal{T}'_j$ is defined as follows:

- if $n_i \in [d]$, it is exactly the same as $\mathcal{T}_i$.
- if $n_i \in [\lambda + s + c]/[d]$, it is the identity.

²³Here, A and each V_i depend on k , but we omit the subscript of k for notational simplicity.

²⁴Here $\prod_j B_j$ means $B_T \cdots B_1$. The order is important because each operation is not commutative in general.

Then, we can prove the following for all $i \in [T]$:

$$\| |\psi_{i-1}\rangle\langle\psi_{i-1}| - |\psi_i\rangle\langle\psi_i| \|_1 \leq O(2^{s+3c/2-d/2}) \quad (126)$$

Since $\text{Tr}_{\mathbf{Z}}[|\psi_0\rangle\langle\psi_0|_{\mathbf{XYZX}'}] = (\mathcal{F}_{k,\mathbf{X} \rightarrow \mathbf{XY}} \otimes \text{id}_{\mathbf{X}'}) (|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|_{\mathbf{XX}'})$, and $\text{Tr}_{\mathbf{Z}}[|\psi_T\rangle\langle\psi_T|_{\mathbf{XYZX}'}] = (\mathcal{I}_{k,\mathbf{X} \rightarrow \mathbf{XY}} \otimes \text{id}_{\mathbf{X}'}) (|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|_{\mathbf{XX}'})$, we obtain Equation (124) from Equation (126). Thus, it remains to prove Equation (126). To this end, we need the following claim.

Claim 6.8. Let A be an isometry which maps ℓ_{in} qubits to ℓ_{out} qubits. Then,

$$(A_{\mathbf{E}} \otimes I_{\mathbf{F}}) |\Omega_{2^{\ell_{\text{in}}}}\rangle_{\mathbf{EF}} = \sqrt{2^{\ell_{\text{out}} - \ell_{\text{in}}}} (I_{\mathbf{E}'} \otimes A_{\mathbf{F}'}^{\top}) |\Omega_{2^{\ell_{\text{out}}}}\rangle_{\mathbf{E}'\mathbf{F}'}, \quad (127)$$

where $\top$ denotes the transpose. Here, $\mathbf{E}$ and $\mathbf{F}$ are ℓ_{in} -qubit registers, and $\mathbf{E}'$ and $\mathbf{F}'$ are ℓ_{out} -qubit registers.

Claim 6.8 follows from a straightforward calculation. We give the proof later. We obtain Equation (126) as follows.

$$\begin{aligned} & \| |\psi_{i-1}\rangle\langle\psi_{i-1}| - |\psi_i\rangle\langle\psi_i| \|_1 \\ &= \left\| \left(\left(\prod_{j=i}^T B_j \mathcal{T}_j' \cdot \prod_{j=1}^{i-1} B_j \mathcal{T}_j \cdot A \right) \otimes \text{id} \right) (|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|) - \left(\left(\prod_{j=i+1}^T B_j \mathcal{T}_j' \cdot \prod_{j=1}^i B_j \mathcal{T}_j \cdot A \right) \otimes \text{id} \right) (|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|) \right\|_1 \\ &= \left\| \left(\left(\prod_{j=i+1}^T B_j \mathcal{T}_j \cdot V_i(\mathcal{T}_i' - \mathcal{T}_i) \cdot \prod_{j=1}^{i-1} B_j \mathcal{T}_j \cdot A \right) \otimes \text{id} \right) (|\Omega_{2\lambda}\rangle\langle\Omega_{2\lambda}|) \right\|_1 \\ &= 2^{s+c} \left\| \left(\left(\prod_{j=i+1}^T B_j \mathcal{T}_j \cdot V_i(\mathcal{T}_i' - \mathcal{T}_i) \right) \otimes \left(\prod_{j=1}^{i-1} B_j \mathcal{T}_j \cdot A \right)^{\top} \right) (|\Omega_{2\lambda+s+c}\rangle\langle\Omega_{2\lambda+s+c}|) \right\|_1 \\ &\quad \text{(By Claim 6.8 with } \ell_{\text{in}} = \lambda \text{ and } \ell_{\text{out}} = \lambda + s + c) \\ &\leq 2^{s+c} \| ((\mathcal{T}_i' - \mathcal{T}_i) \otimes \text{id}) (|\Omega_{2\lambda+s+c}\rangle\langle\Omega_{2\lambda+s+c}|) \|_1 \\ &\quad \text{(By Hölder's inequality Lemma 2.2 with } \|A\|_{\infty} = \|A^{\top}\|_{\infty} \leq 1 \text{ for any isometry } A) \\ &\leq O(2^{s+c3/2-d/2}), \end{aligned} \quad (128)$$

where, in the last line, we have used Lemma 5.6 with $c' = s + c$. which concludes the proof. $\square$

Finally, we give the proof of Claim 6.8.

Proof of Claim 6.8. Let

$$A = \sum_{\substack{x \in \{0,1\}^{\ell_{\text{in}}} \\ y \in \{0,1\}^{\ell_{\text{out}}}}} \alpha_{x,y} |y\rangle\langle x|,$$

then

$$(A_{\mathbf{E}} \otimes I_{\mathbf{F}}) |\Omega_{\ell_{\text{in}}}\rangle_{\mathbf{EF}} = \frac{1}{\sqrt{2^{\ell_{\text{in}}}}} (A_{\mathbf{E}} \otimes I_{\mathbf{F}}) \sum_{x \in \{0,1\}^{\ell_{\text{in}}}} |x\rangle_{\mathbf{E}} |x\rangle_{\mathbf{F}} \quad (129)$$

$$= \frac{1}{\sqrt{2^{\ell_{\text{in}}}}} \sum_{\substack{x \in \{0,1\}^{\ell_{\text{in}}} \\ y \in \{0,1\}^{\ell_{\text{out}}}}} \alpha_{x,y} |y\rangle_{\mathbf{E}} |x\rangle_{\mathbf{F}}, \quad (130)$$

and

$$(I_{\mathbf{E}'} \otimes A_{\mathbf{F}'}^\top) |\Omega_{\ell_{\text{out}}}\rangle_{\mathbf{E}'\mathbf{F}'} = \frac{1}{\sqrt{2^{\ell_{\text{out}}}}} (I_{\mathbf{E}'} \otimes A_{\mathbf{F}'}^\top) \sum_{y \in \{0,1\}^{\ell_{\text{out}}}} |y\rangle_{\mathbf{E}'} |y\rangle_{\mathbf{F}'} \quad (131)$$

$$= \frac{1}{\sqrt{2^{\ell_{\text{out}}}}} \sum_{\substack{x \in \{0,1\}^{\ell_{\text{in}}} \\ y \in \{0,1\}^{\ell_{\text{out}}}}} \alpha_{x,y} |y\rangle_{\mathbf{E}'} |x\rangle_{\mathbf{F}'} . \quad (132)$$

Hence, we have

$$(A_{\mathbf{E}} \otimes I_{\mathbf{F}}) |\Omega_{\ell_{\text{in}}}\rangle_{\mathbf{E}\mathbf{F}} = \sqrt{2^{\ell_{\text{out}} - \ell_{\text{in}}}} (I_{\mathbf{E}'} \otimes A_{\mathbf{F}'}^\top) |\Omega_{\ell_{\text{out}}}\rangle_{\mathbf{E}'\mathbf{F}'} .$$

□

6.2 Proof of Lemma 6.5

In this subsection, we prove Lemma 6.5. To this end, it suffices to apply Lemma 5.9 with $\rho = (\mathcal{F}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)$ and $\xi = (\mathcal{I}_{\lambda \rightarrow \lambda+s,\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)$. Before that, we need to certify that they satisfy the condition in Lemma 5.9. First, we need the following, which corresponds to Lemma 5.7.

Lemma 6.9. *Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be a family of $(\lambda + s + c)$ -qubit unitaries that are QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and with the query access to $\mathcal{U}$, where $\mathcal{U}$ is the UnitaryPSPACE complete problem in Lemma 2.16. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_\lambda| \rceil$. Then, for any polynomial p , there exists a unitary circuit V_λ satisfying the following:*

- V_λ is QPT implementable with classical descriptions of $\mathcal{S}'_n$ for all $n \in [d]$ and with the query access to $\mathcal{U}$.
- V_λ is a $(1, 2^{-p(\lambda)}, \text{poly}(\lambda))$ -block encoding of $(\mathcal{F}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)$.

Since the proof is the same as that of Lemma 5.7, we omit it. Next, we show that $(\mathcal{I}_{\lambda \rightarrow \lambda+s,\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)$ has negligible overlap with the support of $(\mathcal{F}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)$ formalized as follows:

Lemma 6.10. *Suppose that $c(\lambda) = O(\log \lambda)$. Let Q' be the projection onto the support of $(\mathcal{F}_{\{V_k\},\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)$. Then,*

$$\text{Tr}[Q'(\mathcal{I}_{\lambda \rightarrow \lambda+s,\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|)] \leq \text{negl}(\lambda). \quad (133)$$

We need the following lemma for the proof of Lemma 6.10.

Lemma 6.11. *Let $\lambda, s, \ell \in \mathbb{N}$ such that $2^{\lambda+s} \geq \ell^2$. Then, we have*

$$\left\| (\mathcal{I}_{\lambda \rightarrow \lambda+s,\ell} \otimes \text{id})(|\Omega_{2^{\lambda\ell}}\rangle\langle\Omega_{2^{\lambda\ell}}|) - \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{2^{\lambda+s}}} |\psi\rangle\langle\psi|^{\otimes \ell} \right\|_1 \leq O\left(\frac{\ell^2}{2^{\lambda+s}}\right), \quad (134)$$

where $\mathcal{I}_{\lambda \rightarrow \lambda+s,\ell}$ is the Haar random isometry map defined in Definition 6.4

Before proving Lemma 6.11, we give the proof of Lemma 6.10 assuming Lemma 6.11.

Proof of Lemma 6.10. First, we prove that $\text{Tr}[Q'] \leq 2^{(1+c)\ell}$. Note that

$$\begin{aligned} & (\mathcal{F}_{\{V_k\}, \ell, \mathbf{A} \rightarrow \mathbf{A}\mathbf{B}} \otimes \text{id}_{\mathbf{A}'}) (|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})) \\ &= \frac{1}{|\mathcal{K}_\lambda|} \sum_{k \in \mathcal{K}_\lambda} \text{Tr}_{\mathbf{C}}[(V_{k, \mathbf{C}\mathbf{B}\mathbf{A}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'}) (|0^c\rangle\langle 0^c|_{\mathbf{C}}^{\otimes \ell} \otimes |0^s\rangle\langle 0^s|_{\mathbf{B}}^{\otimes \ell} |\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})). \end{aligned} \quad (135)$$

For each k , the rank of $\text{Tr}_{\mathbf{C}}[(V_{k, \mathbf{C}\mathbf{B}\mathbf{A}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'}) (|0^c\rangle\langle 0^c|_{\mathbf{C}}^{\otimes \ell} \otimes |0^s\rangle\langle 0^s|_{\mathbf{B}}^{\otimes \ell} |\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})]$ is at most $\min\{2^{c\ell}, 2^{(2\lambda+s)\ell}\} = 2^{c\ell}$ since $(V_{k, \mathbf{C}\mathbf{B}\mathbf{A}}^{\otimes \ell} \otimes \text{id}_{\mathbf{A}'}) (|0^c\rangle\langle 0^c|_{\mathbf{C}}^{\otimes \ell} \otimes |0^s\rangle\langle 0^s|_{\mathbf{B}}^{\otimes \ell} |\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{A}\mathbf{A}'})$ is pure. Thus, the rank of Q' is at most $2^{c\ell} \cdot |\mathcal{K}_\lambda| \leq 2^{(1+c)\ell}$, which implies $\text{Tr}[Q'] \leq 2^{(1+c)\ell}$.

Note that the rank of Q' is at most 2^ℓ since it is the same as the rank of $(\mathcal{M}_{\{\mathcal{I}_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$. Therefore, we have

$$\begin{aligned} \text{Tr}[Q'(\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)] &\leq \text{Tr}\left[Q' \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{2^{2\lambda+s}}}} |\psi\rangle\langle\psi|^{\otimes \ell}\right] + \text{negl}(\lambda) && \text{(By Lemma 6.11)} \\ &= \frac{\text{Tr}[Q' \Pi_{\text{sym}}]}{\binom{2^{2\lambda+s} + \ell - 1}{\ell}} + \text{negl}(\lambda) && \text{(By Lemma 2.10)} \\ &\leq \frac{2^{(1+c)\ell}}{\binom{2^{2\lambda+s} + \ell - 1}{\ell}} + \text{negl}(\lambda) && \text{(By } \text{Tr}[Q' \Pi_{\text{sym}}] \leq \text{Tr}[Q'] \leq 2^{(1+c)\ell})} \\ &\leq O\left(\frac{2^{(1+c)\ell}(\ell!)}{2^{(2\lambda+s)\ell}}\right) + \text{negl}(\lambda) \\ &\leq O\left(\frac{2^{(1+c)\ell} \ell^{\ell+1/2} e^{-\ell}}{2^{(2\lambda+s)\ell}}\right) + \text{negl}(\lambda) && \text{(By the Stirling's formula)} \\ &= O\left(\ell^{1/2} \left(\frac{2^c e^{-1} \ell}{2^{2\lambda+s}}\right)^\ell\right) + \text{negl}(\lambda) \\ &\leq O\left(\ell^{1/2} \left(\frac{\text{poly}(\lambda)}{2^{2\lambda+s}}\right)^\ell\right) + \text{negl}(\lambda) && \text{(By } c(\lambda) = O(\log \lambda)) \\ &\leq \text{negl}(\lambda), \end{aligned} \quad (136)$$

which concludes the proof. $\square$

For the proof of Lemma 6.11, we use the following.

Lemma 6.12 (Haar Twirl Approximation, [SHH24, HY24]). *Let $n, \ell \in \mathbb{N}$ such that $2^n \geq \ell^2$. Let $\mathbf{A}$ be a $n\ell$ -qubit register, and $\mathbf{A}'$ be some fixed register. Then, for any quantum state ρ on the registers $\mathbf{A}\mathbf{A}'$,*

$$\left\| (\mathcal{M}_{\mu_{2^n, \ell, \mathbf{A}}} \otimes \text{id}_{\mathbf{A}'}) (\rho_{\mathbf{A}\mathbf{A}'})) - \sum_{\pi \in S_\ell} \frac{1}{2^{n\ell}} R_{\pi, \mathbf{A}} \otimes \text{Tr}_{\mathbf{A}'}[(R_{\pi, \mathbf{A}}^\dagger \otimes I_{\mathbf{A}'})\rho_{\mathbf{A}\mathbf{A}'}] \right\|_1 \leq O\left(\frac{\ell^2}{2^n}\right). \quad (137)$$

Here, S_ℓ denotes the permutation group over ℓ elements, and for $\pi \in S_\ell$, R_π is the permutation unitary such that $R_\pi |x_1, \dots, x_\ell\rangle = |x_{\pi^{-1}(1)}, \dots, x_{\pi^{-1}(\ell)}\rangle$ for all $x_1, \dots, x_\ell \in \{0, 1\}^n$.

From Lemma 6.12 and a straightforward calculation, we can prove Lemma 6.11.

Proof of Lemma 6.11. We define $\mathbf{A}$ and $\mathbf{A}'$ are $\lambda\ell$ -qubit register, and $\mathbf{B}$ is $s\ell$ -qubit register. Let the output registers of $\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell, \mathbf{A}}$ be $\mathbf{A}$ and $\mathbf{B}$. We prove Lemma 6.11 via the following hybrids:

$$\rho_{0, \mathbf{BAA}'} := (\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell, \mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'} \quad (138)$$

$$\rho_{1, \mathbf{BAA}'} := \sum_{\pi \in S_\ell} \frac{1}{2^{(\lambda+s)\ell}} R_{\pi, \mathbf{BA}} \otimes \text{Tr}_{\mathbf{BA}}[(R_{\pi, \mathbf{BA}}^\dagger \otimes I_{\mathbf{A}'})(|0^{s\ell}\rangle\langle 0^{s\ell}|_{\mathbf{B}} \otimes |\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'}]) \quad (139)$$

$$\rho_{2, \mathbf{BAA}'} := \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{2\lambda+s}} |\psi\rangle\langle\psi|^{\otimes \ell}_{\mathbf{BAA}'}. \quad (140)$$

Since we have $\|\rho_0 - \rho_1\|_1 \leq O(\ell^2/2^{\lambda+s})$ from Lemma 6.12 with $n = \lambda + s$, it suffices to prove $\|\rho_1 - \rho_2\|_1 \leq O(\ell^2/2^{\lambda+s})$. Note that we have

$$\begin{aligned} & (R_{\pi, \mathbf{BA}}^\dagger \otimes I_{\mathbf{A}'})(|0^{s\ell}\rangle\langle 0^{s\ell}|_{\mathbf{B}} \otimes |\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'}) \\ &= R_{\pi, \mathbf{B}}^\dagger |0^{s\ell}\rangle\langle 0^{s\ell}|_{\mathbf{B}} \otimes (R_{\pi, \mathbf{A}}^\dagger \otimes I_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'}) \quad (R_{\pi, \mathbf{BA}}^\dagger = R_{\pi, \mathbf{B}}^\dagger \otimes R_{\pi, \mathbf{A}}^\dagger) \\ &= |0^{s\ell}\rangle\langle 0^{s\ell}|_{\mathbf{B}} \otimes (R_{\pi, \mathbf{A}}^\dagger \otimes I_{\mathbf{A}'})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'}) \quad ((0^s, \dots, 0^s) \text{ is invariant under any permutation } \pi \in S_\ell) \\ &= |0^{s\ell}\rangle\langle 0^{s\ell}|_{\mathbf{B}} \otimes (I_{\mathbf{A}} \otimes R_{\pi, \mathbf{A}'})|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'}. \end{aligned} \quad (141)$$

In the last line, we used the fact that $R_{\pi}^\dagger = R_{\pi}^\top$ for any $\pi \in S_\ell$, where $\top$ denotes the transpose. Thus, we have

$$\begin{aligned} & R_{\pi, \mathbf{BA}} \otimes \text{Tr}_{\mathbf{BA}}[(R_{\pi, \mathbf{BA}}^\dagger \otimes I_{\mathbf{B}})(|0^{s\ell}\rangle\langle 0^{s\ell}|_{\mathbf{B}} \otimes |\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'})] \\ &= R_{\pi, \mathbf{BA}} \otimes \text{Tr}_{\mathbf{A}}[(I_{\mathbf{A}} \otimes R_{\pi, \mathbf{A}'})|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|_{\mathbf{AA}'}] \quad (\text{By Equation (141)}) \\ &= \frac{1}{2^{\lambda\ell}} R_{\pi, \mathbf{BA}} \otimes R_{\pi, \mathbf{A}'} \\ &= \frac{1}{2^{\lambda\ell}} R_{\pi, \mathbf{BAA}'}, \end{aligned} \quad (142)$$

where we have used $R_{\pi, \mathbf{BA}} \otimes R_{\pi, \mathbf{A}'} = R_{\pi, \mathbf{BAA}'}$ in the last line. From Equation (142) and Lemma 2.10, we have

$$\rho_{1, \mathbf{BAA}'} = \sum_{\pi \in S_\ell} \frac{1}{2^{(2\lambda+s)\ell}} R_{\pi, \mathbf{BAA}'} = \frac{\ell!}{2^{(2\lambda+s)\ell}} \binom{2^{\lambda+s} + \ell - 1}{\ell} \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{2\lambda+s}} |\psi\rangle\langle\psi|^{\otimes \ell}_{\mathbf{BAA}'}. \quad (143)$$

By a straightforward calculation, we have $\frac{\ell!}{2^{(2\lambda+s)\ell}} \binom{2^{\lambda+s} + \ell - 1}{\ell} = 1 + O(\ell^2/2^{2\lambda+s})$. Therefore, since $\rho_2 = \mathbb{E}_{|\psi\rangle \leftarrow \sigma_{2\lambda+s}} |\psi\rangle\langle\psi|^{\otimes \ell}$, we obtain $\|\rho_1 - \rho_2\|_1 \leq O(\ell^2/2^{2\lambda+s}) \leq O(\ell^2/2^{\lambda+s})$, which concludes the proof. $\square$

From Lemmata 6.9 and 6.10, we are ready to prove Lemma 6.5. We restate it here.

Lemma 6.5. Suppose that $c(\lambda) = O(\log \lambda)$. Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be the family of $(\lambda + s + c)$ -qubit unitary, and $\mathcal{F}_{\{V_k\}_{k \in \mathcal{K}_\lambda}, \ell}$ be the CPTP map in Lemma 6.3. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_\lambda| \rceil$. Then, there exists a QPT algorithm $\mathcal{D}^\mathcal{U}$ that, on input classical descriptions of S'_n for all $n \in [d]$, distinguishes $(\mathcal{F}_{\{V_k\}_{k \in \mathcal{K}_\lambda}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ from $(\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ with advantage at least $1 - \text{negl}(\lambda)$.

Proof of Lemma 6.5. Let $\rho = (\mathcal{F}_{\{V_k\}_{k \in \mathcal{K}_\lambda}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$. From Lemmata 6.9 and 6.10, they satisfy the condition in Lemma 5.9. Therefore, we obtain Lemma 6.5 by applying Lemma 5.9, which concludes the proof. $\square$

7 Oracle Separation Between PRIs with Short Stretch and PRIs with Large Stretch

In this section, we prove the following theorem.

Theorem 7.1. *Let $s(\lambda) = O(\log \lambda)$ and $t(\lambda) = \Omega(\lambda)$. Then, there exists a unitary oracle $\mathcal{O}'$ relative to which*

- *adaptive PRIs with t stretch exist, but*
- *non-adaptive and ancilla-free PRIs with s stretch do not exist.*

Here, it is allowed to query the inverse of $\mathcal{O}'$.

Since PRIs with $s = 0$ stretch are PRUs, we get the following theorem immediately by choosing $s = 0$.

Theorem 7.2. *Let $t(\lambda) = \Omega(\lambda)$. Then, there exists a unitary oracle $\mathcal{O}'$ relative to which*

- *adaptive PRIs with t stretch exist, but*
- *non-adaptive and ancilla-free PRUs do not exist.*

Here, it is allowed to query the inverse of $\mathcal{O}'$.

7.1 Separation Oracle

We define a separation oracle.

Definition 7.3 (Haar Random Isometry Unitary Oracle). *For a function $t : \mathbb{N} \rightarrow \mathbb{N}$, we define a unitary oracle HRI_t as follows: For each $n \in \mathbb{N}$ and $m \in \{0, 1\}^n$, sample $(n+t(n))$ -qubit unitary $U_{n,m} \in \mathbb{U}(2^{n+t(n)})$ from the Haar measure $\mu_{2^{n+t(n)}}$. Then, define the $(n+t(n)+1)$ -qubit unitary*

$$\text{HRI}_{t,n,m} := |0\rangle\langle 1| \otimes (|0^t\rangle\langle 0^t| \otimes I_n) U_{n,m}^\dagger + |1\rangle\langle 0| \otimes U_{n,m} (|0^t\rangle\langle 0^t| \otimes I_n) + I_\perp^{t,n,m}, \quad (144)$$

where, $I_\perp^{t,n,m}$ is the identity on the subspace orthogonal to $\text{span}\{|0\rangle \otimes |0^t\rangle |x\rangle, |1\rangle \otimes U_{n,m}(|0^t\rangle |x\rangle)\}_{x \in \{0,1\}^n}$. We define $\text{HRI}_t := \{\text{HRI}_{t,n}\}_{n \in \mathbb{N}}$, where $\text{HRI}_{t,n} := \{\text{HRI}_{t,n,m}\}_{m \in \{0,1\}^n}$.

Remark 7.4. We have $\text{HRI}_{t,n,m} = \text{HRI}_{t,n,m}^\dagger$ for any function t , $n \in \mathbb{N}$ and $m \in \{0, 1\}^n$ regardless of the choice of $U_{n,m}$.

Next, we construct a PRI with t stretch relative to this oracle.

Definition 7.5 (PRI Relative to HRI Oracle). *Let $t : \mathbb{N} \rightarrow \mathbb{N}$ be a function. We define a QPT algorithm G^{HRI_t} as follows:*

1. *Let $k \in \{0, 1\}^\lambda$ and λ -qubit register $\mathbf{A}$ be inputs.*
2. *Prepare $|0\rangle_{\mathbf{X}} |0^t\rangle_{\mathbf{Y}}$ on ancilla register $\mathbf{XY}$.*
3. *Apply $I_{\mathbf{X}} \otimes U_{\lambda,k\mathbf{YA}}$ by querying the registers $\mathbf{X}$, $\mathbf{Y}$ and $\mathbf{A}$ to $\text{HRI}_{t,\lambda,k}$.*
4. *Output the registers $\mathbf{Y}$ and $\mathbf{A}$.*

Remark 7.6. If the input state is a pure state $|\psi\rangle_{\mathbf{A}}$, the output state is $G^{\text{HRI}_t}(k, |\psi\rangle_{\mathbf{A}}) = U_{\lambda,k\mathbf{YA}}(|0^t\rangle_{\mathbf{Y}} |\psi\rangle_{\mathbf{A}})$.

Our goal is to prove the following theorem, which implies Theorem 7.1.

Theorem 7.7. *Let $\mathcal{U}$ be a **UnitaryPSPACE** complete problem in Lemma 2.16. Let $s(\lambda) = O(\log \lambda)$ and $t(\lambda) = \Omega(\lambda)$. Then, with probability 1 over the choice of HRI_t defined in Definition 7.3, the following are satisfied:*

1. G^{HRI_t} in Definition 7.5 is an adaptive secure PRI with t stretch relative to $(\text{HRI}_t, \mathcal{U})$
2. Non-adaptive PRIs with s stretch do not exist relative to $(\text{HRI}_t, \mathcal{U})$.

Since the proof strategy of the first item in Theorem 7.7 is the same as Theorem 4.2, we omit it.

7.2 Breaking PRIs with Short Stretch

In this subsection, we prove the second item in Theorem 7.7. We give an adversary in a similar way as in Algorithms 1 and 2. Suppose that $\mathcal{U}$ is the **UnitaryPSPACE**-complete problem. Let $G^{\text{HRI}_t, \mathcal{U}}$ be a QPT algorithm that satisfies the correctness of ancilla-free PRIs with s stretch. For such $G^{\text{HRI}_t, \mathcal{U}}$, let $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ be the isometry implemented by $G^{\text{HRI}_t, \mathcal{U}}$ on input $k \in \mathcal{K}_\lambda$, where $\mathcal{K}_\lambda$ denotes the key-space. We define the following map:

$$\mathcal{M}_{\{\mathcal{I}_k\}, \ell}(\cdot) = \mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} \mathcal{I}_k^{\otimes \ell}(\cdot) \mathcal{I}_k^{\dagger \otimes \ell}. \quad (145)$$

To construct a PRI adversary, we need two lemmas as in Sections 5 and 6. These lemmas can be obtained by modifying Lemma 6.3 and Lemma 6.5 for the HRI_t oracle. We give their proofs later.

Lemma 7.8. *Let $T(\lambda)$ be a polynomial. Let $\epsilon > 0$ and $d \in \mathbb{N}$. Let $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ be an ensemble of isometries mapping λ qubits to $\lambda + s$ qubits, where each $\mathcal{I}_k$ is QPT implementable with s ancilla qubits by making T queries to $(\text{HRI}_t, \mathcal{U})$. For all $n \in [d]$ and $m \in \{0, 1\}^n$, let $\text{HRI}'_{t, n, m}$ be any unitary satisfying*

$$\|\text{HRI}_{t, n, m}(\cdot) \text{HRI}_{t, n, m}^\dagger - \text{HRI}'_{t, n, m}(\cdot) \text{HRI}'_{t, n, m}{}^\dagger\|_\diamond \leq \epsilon. \quad (146)$$

Then, for any polynomial ℓ , there exists a family $\{V_k\}_{k \in \mathcal{K}_\lambda}$ of $(\lambda + s + c)$ -qubit unitaries such that each V_k is QPT implementable with classical descriptions of $\text{HRI}'_{t, n, m}$ for all $n \in [d]$, $m \in \{0, 1\}^n$ and query access to $\mathcal{U}$ such that it satisfies

$$\left\| (\mathcal{M}_{\{\mathcal{I}_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) - (\mathcal{F}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|) \right\|_1 \leq O(\ell T \epsilon) + O\left(\frac{2^{s+3c/2} \ell T}{2^{t(d)/2}}\right). \quad (147)$$

Here, $\mathcal{F}_{\{V_k\}, \ell}$ is a CPTP map from $\lambda\ell$ qubits to $(\lambda + s)\ell$ qubits defined as follows:

$$\mathcal{F}_{\{V_k\}, \ell}((\cdot)_\mathbf{A}) := \text{Tr}_\mathbf{C} \left[\mathbb{E}_{k \leftarrow \mathcal{K}_\lambda} V_{k, \mathbf{ABC}}^{\otimes \ell} ((\cdot)_\mathbf{A} \otimes |0^s\rangle\langle 0^s|_\mathbf{B}^{\otimes \ell} \otimes |0^c\rangle\langle 0^c|_\mathbf{C}^{\otimes \ell}) V_{k, \mathbf{ABC}}^{\dagger \otimes \ell} \right], \quad (148)$$

where $\mathbf{A}$ is a $\lambda\ell$ -qubit register, $\mathbf{B}$ is a $s\ell$ -qubit register, and $\mathbf{C}$ is a $c\ell$ -qubit register.

Regarding the next lemma, recall that $\mathcal{I}_{\lambda \rightarrow \lambda+s}$ is the Haar random isometry map defined in Definition 6.4.

Lemma 7.9. *Suppose that $c(\lambda) = O(\log \lambda)$. Let $\{V_k\}_{k \in \mathcal{K}_\lambda}$ be a family of $(\lambda + s + c)$ -qubit unitaries, and $\mathcal{F}_{\{V_k\}, \ell}$ be the CPTP map in Lemma 7.8. Let $\ell(\lambda) := \lceil \log |\mathcal{K}_\lambda| \rceil$. Then, there exists a QPT algorithm $\mathcal{D}^\mathcal{U}$ that, on input classical descriptions of $\text{HRI}'_{t, n, m}$ for all $n \in [d]$ and $m \in \{0, 1\}^n$, distinguishes $(\mathcal{F}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ from $(\mathcal{I}_{\lambda \rightarrow \lambda+s, \ell} \otimes \text{id})(|\Omega_{2\lambda\ell}\rangle\langle\Omega_{2\lambda\ell}|)$ with advantage at least $1 - \text{negl}(\lambda)$.*

Algorithm 3 Adversary distinguishing $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ from Haar random isometry relative to $(\text{HRI}, \mathcal{U})$.

Oracle access: The algorithm has query access to

- an isometry $\mathcal{I}$ from λ qubits to $\lambda + s(\lambda)$ qubits, which is whether $\mathcal{I}_k$ or a Haar random isometry.
- the oracle $\mathcal{O}' = (\text{HRI}_t, \mathcal{U})$ and its inverse, where HRI_t is defined in Definition 7.3.

Input: The algorithm takes the security parameter 1^λ as input.

Define $\ell := \lceil \log |\mathcal{K}_\lambda| \rceil$ and $d := (2 \log(\ell T p) + 2s + 3c)^{\frac{1}{a}}$ where p is a polynomial, and T is the number of queries to $\mathcal{O}'$ to implement $\mathcal{I}_k$.

1. For $n \in [d]$ and $m \in \{0, 1\}^n$, run the process tomography algorithm in Theorem 2.7 on inputs $\epsilon := \frac{1}{\ell T p}$ and $\eta := 2^{-\lambda-1}$ for $\text{HRI}_{t,n,m}$ to get a classical description of $\text{HRI}'_{t,n,m}$. Note that $\|\text{HRI}_{t,n,m}(\cdot) \text{HRI}_{t,n,m}^\dagger - \text{HRI}'_{t,n,m}(\cdot) \text{HRI}_{t,n,m}^{\dagger'}\|_\diamond \leq \epsilon$ holds with probability at least $1 - 2^{-\lambda}$ over the randomness of the process tomography algorithm.
2. Prepare $(\mathcal{I}^{\otimes \ell} \otimes I) |\Omega_{2\lambda\ell}\rangle$ by querying $\mathcal{I}$.
3. Let $\{\mathcal{I}'_k\}_{k \in \mathcal{K}_\lambda}$ be a family of isometries in Lemma 7.8. Note that each $\mathcal{I}'_k$ is QPT implementable with access to $\mathcal{U}$ and classical descriptions of $\text{HRI}'_{t,n,m}$ for all $n \in [d]$ and $m \in \{0, 1\}^n$, where such classical descriptions are obtained in the step 1. Let $\mathcal{D}^{(\cdot)}$ be a QPT algorithm in Lemma 7.9 for $\{\mathcal{I}'_k\}_{k \in \mathcal{K}_\lambda}$. By querying $\mathcal{U}$, run $\mathcal{D}^\mathcal{U}$ on input $(\mathcal{I}^{\otimes \ell} \otimes I) |\Omega_{2\lambda\ell}\rangle$ and classical descriptions of $\text{HRI}'_{t,n,m}$ for all $n \in [d]$ and $m \in \{0, 1\}^n$ to get $b \in \{0, 1\}$.

Output: The algorithm outputs b .

From these lemmas, we construct a PRI adversary as shown in Algorithm 3. The red-highlighted lines in Algorithm 3 indicate the differences from Algorithm 2.

Remark 7.10. d is chosen so that it satisfies $\ell T 2^{s+3c/2-t(d)/2} = O(1/p)$ and ensures that the process tomography algorithm runs in QPT. Algorithm 3 is QPT because, if $n \in [d]$, the dimension of $\text{HRI}_{t,n,m}$ is at most $2^{d+t(d)+1} = O(2^{2t(d)}) = O((\ell T p)^4 \cdot 2^{4s+6c}) \leq \text{poly}(\lambda)$, where the inequality follows from $s(\lambda) \leq O(\log \lambda)$ and $c(\lambda) \leq O(\log \lambda)$.

Now we are ready to construct the PRI adversary in Algorithm 3. The following Theorem 7.11 implies the second statement in Theorem 7.7.

Theorem 7.11. *Suppose that $s = O(\log \lambda)$ and $t(\lambda) = \Theta(\lambda^a)$ for some constant $a \geq 1$. Let $\mathcal{O}' := (\text{HRI}_t, \mathcal{U})$. Let $G^{\mathcal{O}'}$ be a QPT algorithm that satisfies the correctness of ancilla-free PRIs with s stretch. For such $G^{\mathcal{O}'}$, let $\{\mathcal{I}_k\}_{k \in \mathcal{K}_\lambda}$ be the isometry mapping λ qubits to $\lambda + s$ qubits implemented by $G^{\mathcal{O}'}$ on input $k \in \mathcal{K}_\lambda$, where $\mathcal{K}_\lambda$ denotes the key-space. Then, for any polynomial p , the QPT adversary $\mathcal{A}^{(\cdot, \cdot)}$ defined in Algorithm 3 satisfies*

$$\left| \Pr_{k \leftarrow \mathcal{K}_\lambda} [1 \leftarrow \mathcal{A}^{\mathcal{I}_k, \mathcal{O}'}(1^\lambda)] - \Pr_{U \leftarrow \mu_{2^{\lambda+s}}} [1 \leftarrow \mathcal{A}^{\mathcal{I}_U, \mathcal{O}'}(1^\lambda)] \right| \geq 1 - O\left(\frac{1}{p(\lambda)}\right), \quad (149)$$

where, for each $U \in \mathbb{U}(2^{\lambda+s(\lambda)})$, $\mathcal{I}_U$ is the isometry that maps λ -qubit state $|\psi\rangle$ to $(\lambda + s(\lambda))$ -qubit state $U(|\psi\rangle |0^s\rangle)$. Here, $\mathcal{A}^{(\cdot, \cdot), \mathcal{O}'}$ queries not only $\mathcal{O}'$ but also its inverse. Moreover, $\mathcal{A}^{(\cdot, \cdot), \mathcal{O}'}$ queries the first oracle non-adaptively.

7.3 Proof of Lemma 7.8

To conclude the proof, it remains to prove Lemmata 7.8 and 7.9. Since the proof of Lemma 7.9 is the same as that of Lemma 6.5, we omit it. Thus, it suffices to prove Lemma 7.8.

The proof strategy is the same as that of Lemmata 5.1 and 6.3. First, we need the following lemma.

Lemma 7.12. *Let $n + t(n) + 1 \leq \lambda + c'$. Suppose that $\mathbf{A}$ and $\mathbf{A}'$ are λ -qubit registers, and $\mathbf{B}$ is an s -qubit register. Then, for any $U, V \in \mathbb{U}(2^{\lambda+c'})$ and $m \in \{0, 1\}^n$,*

$$\frac{1}{2} \| |\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| \|_1 \leq O(2^{c'-t(n)/2}), \quad (150)$$

where

$$|\psi\rangle_{\mathbf{BAA}'} := ((U(\text{HRI}_{t,n,m} \otimes I)V)_{\mathbf{BA}} \otimes I_{\mathbf{A}'} |0^s\rangle_{\mathbf{B}} |\Omega_{2^\lambda}\rangle_{\mathbf{AA}'}), \quad (151)$$

$$|\phi\rangle_{\mathbf{BAA}'} := ((UV)_{\mathbf{BA}} \otimes I_{\mathbf{A}'} |0^s\rangle_{\mathbf{B}} |\Omega_{2^\lambda}\rangle_{\mathbf{AA}'}), \quad (152)$$

and $\text{HRI}_{t,n,m}$ is defined in Definition 7.3.

Proof of Lemma 7.12. First, we can see $(\text{HRI}_{t,n,m} \otimes I^{\otimes 2\lambda+2c'-n-t(n)-1}) |\Omega_{2^{\lambda+c'}}\rangle$ is close to $|\Omega_{2^{\lambda+c'}}\rangle$ in the

trace norm as follows:

$$\begin{aligned}
& \frac{1}{2} \left\| (\text{HRI}_{t,n,m} \otimes I^{\otimes 2\lambda+2c'-n-t(n)-1}) |\Omega_{2\lambda+c'}\rangle \langle \Omega_{2\lambda+c'}| (\text{HRI}_{t,n,m} \otimes I^{\otimes 2\lambda+2c'-n-t(n)-1})^\dagger - |\Omega_{2\lambda+c'}\rangle \langle \Omega_{2\lambda+c'}| \right\|_1 \\
&= \sqrt{1 - |\langle \Omega_{2\lambda+c'} | (\text{HRI}_{t,n,m} \otimes I^{\otimes 2\lambda+2c'-n-t(n)-1}) | \Omega_{2\lambda+c'} \rangle|^2} \\
& \quad \text{(By } \frac{1}{2} \| |\alpha\rangle\langle\alpha| - |\beta\rangle\langle\beta| \|_1 = \sqrt{1 - |\langle\alpha|\beta\rangle|^2} \text{)} \\
&= \sqrt{1 - \frac{1}{2^{2\lambda+2c'}} |\text{Tr}[\text{HRI}_{t,n,m} \otimes I^{\otimes \lambda+c'-n-t(n)-1}]|^2} \\
&= \sqrt{1 - \frac{1}{2^{2n+2t(n)+2}} |\text{Tr}[\text{HRI}_{t,n,m}]|^2} \quad \text{(By } \text{Tr}[A \otimes B] = \text{Tr}[A]\text{Tr}[B] \text{)} \\
&= \sqrt{1 - \frac{(2^{n+t(n)+1} - 2^{n+1})^2}{2^{2n+2t(n)+2}}} \quad \text{(By } \text{Tr}[\text{HRI}_{t,n,m}] = 2^{n+t(n)+1} - 2^{n+1} \text{)} \\
&\leq O\left(\frac{1}{2^{t(n)/2}}\right). \tag{153}
\end{aligned}$$

Let us define an isometry $V'_{\mathbf{BA}} := V_{\mathbf{BA}}(|0^s\rangle_{\mathbf{B}} \otimes I_{\mathbf{A}})$ mapping $\mathbf{A}$ to $\mathbf{BA}$. With this V' , we have

$$\begin{aligned}
|\psi\rangle_{\mathbf{BA}\mathbf{A}'} &= ((U(\text{HRI}_{t,n,m} \otimes I))_{\mathbf{BA}} \cdot V'_{\mathbf{A}} \otimes I_{\mathbf{A}'}) |\Omega_{2\lambda}\rangle_{\mathbf{AA}'} \\
|\phi\rangle_{\mathbf{BA}\mathbf{A}'} &= (U_{\mathbf{BA}} \cdot V'_{\mathbf{A}} \otimes I_{\mathbf{A}'}) |\Omega_{2\lambda}\rangle_{\mathbf{AA}'} \tag{154}
\end{aligned}$$

Then, from Equations (153) and (154), we obtain our statement as follows:

$$\begin{aligned}
& \frac{1}{2} \| |\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| \|_1 \\
&= \frac{1}{2} \left\| ((U(\text{HRI}_{t,n,m} \otimes I))_{\mathbf{BA}} \cdot V'_{\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(\Omega_{2\lambda, \mathbf{AA}'}) - (U_{\mathbf{BA}} \cdot V'_{\mathbf{A}} \otimes \text{id}_{\mathbf{A}'})(\Omega_{2\lambda, \mathbf{AA}'}) \right\|_1 \\
& \quad \text{(By Equation (154))} \\
&= \frac{2^{c'}}{2} \left\| ((U(\text{HRI}_{t,n,m} \otimes I))_{\mathbf{BA}} \otimes V_{\mathbf{B}'\mathbf{A}'}^\top)(\Omega_{2\lambda+c', \mathbf{BAB}'\mathbf{A}'}) - (U_{\mathbf{BA}} \otimes V_{\mathbf{B}'\mathbf{A}'}^\top)(\Omega_{2\lambda+c', \mathbf{BAB}'\mathbf{A}'}) \right\|_1 \\
& \quad \text{(By Claim 6.8 and define } \mathbf{B}' \text{ as an } c' \text{-qubit register)} \\
&= \frac{2^{c'}}{2} \left\| ((\text{HRI}_{t,n,m} \otimes I)_{\mathbf{BA}} \otimes \text{id}_{\mathbf{B}'\mathbf{A}'})(\Omega_{2\lambda+c', \mathbf{BAB}'\mathbf{A}'}) - (\Omega_{2\lambda+c', \mathbf{BAB}'\mathbf{A}'}) \right\|_1 \\
& \quad \text{(By Hölder's inequality (Lemma 2.2) and } \|V'\|_\infty = \|V'^\top\|_\infty \leq 1 \text{ since } V' \text{ is an isometry)} \\
&\leq O(2^{c'-t(n)/2}), \tag{By Equation (153)}
\end{aligned}$$

where $V'^\top$ denotes the transpose of V' . □

Now we are ready to prove Lemma 7.8.

Proof of Lemma 7.8. The proof strategy is the same as in Lemma 6.3. The difference lies in applying Lemma 7.12 instead of Lemma 5.6. Thus, it suffices to replace the dependence on d with the dependence on $t(d)$. Therefore, there exists a family $\{V_k\}_{k \in \mathcal{K}_\lambda}$ of $(\lambda + s + c)$ -qubit unitaries such that each V_k is QPT implementable with classical descriptions of $\text{HRI}'_{t,n,m}$ for all $n \in [d]$ and $m \in \{0, 1\}^n$, and query access to

the **UnitaryPSPACE**-complete oracle $\mathcal{U}$ such that it satisfies

$$\left\| (\mathcal{M}_{\{\mathcal{I}_k\}, \ell} \otimes \text{id})(|\Omega_{2^{\lambda \ell}}\rangle\langle\Omega_{2^{\lambda \ell}}|) - (\mathcal{F}_{\{V_k\}, \ell} \otimes \text{id})(|\Omega_{2^{\lambda \ell}}\rangle\langle\Omega_{2^{\lambda \ell}}|) \right\|_1 \leq O(\ell T \epsilon) + O\left(\frac{2^{s+3c/2} \ell T}{2^{t(d)/2}}\right), \quad (155)$$

which concludes the proof. $\square$

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A Relationship Between Black-Box Construction and Oracle Separation

A.1 Impossibility of Black-Box Constructions from Oracle Separations

[CM24, CCS24] showed the relation between oracle separations and black-box constructions. Therefore, by applying the proof for Theorem 1.7, we obtain Theorem 1.1. Here for the convenience of readers, we provide the proof.

Theorem 1.1. *There is no black-box construction of non-adaptive and $O(\log \lambda)$ -ancilla PRUs from PRFSGs.*

Proof of Theorem 1.1. For the sake of contradiction, assume that there is a black-box construction of PRUs from PRFSGs. Then there exist QPT algorithms $C^{(\cdot, \cdot)}$ and $R^{(\cdot, \cdot)}$ such that

1. Black-box construction: For any PRFSG G and any its unitary implementation $\tilde{G}$,²⁵ $C^{\tilde{G}, \tilde{G}^\dagger}$ satisfies correctness of non-adaptive PRUs.
2. Black-box security reduction: For any PRFSG G , any its unitary implementation $\tilde{G}$, any adversary $\mathcal{A}$ that breaks the security of $C^{\tilde{G}, \tilde{G}^\dagger}$, and any unitary implementation $\tilde{A}$ of $\mathcal{A}$, it holds that $R^{\tilde{A}, \tilde{A}^\dagger}$ breaks the security of G .

From Theorem 1.7, there is a QPT algorithm $B^{\mathcal{O}, \mathcal{O}^\dagger}$ querying $\mathcal{O}$ and $\mathcal{O}^\dagger$ such that $B^{\mathcal{O}, \mathcal{O}^\dagger}$ is a PRFSGs. Therefore $C^{\tilde{B}^{\mathcal{O}, \mathcal{O}^\dagger}, (\tilde{B}^{\mathcal{O}, \mathcal{O}^\dagger})^\dagger}$ satisfies the correctness of non-adaptive PRUs. This means that a QPT algorithm $D^{\mathcal{O}, \mathcal{O}^\dagger}$ querying $\mathcal{O}$ and $\mathcal{O}^\dagger$ satisfies correctness of non-adaptive PRU. However, because non-adaptive PRUs do not exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$ from Theorem 1.7, this should not be secure. Therefore there exists a QPT adversary $\mathcal{A}^{\mathcal{O}, \mathcal{O}^\dagger}$ that breaks it. Then $R^{\tilde{\mathcal{A}}^{\mathcal{O}, \mathcal{O}^\dagger}, \tilde{\mathcal{A}}^{\mathcal{O}, \mathcal{O}^\dagger}}$ breaks $B^{\mathcal{O}, \mathcal{O}^\dagger}$, which means that a QPT algorithm $E^{\mathcal{O}, \mathcal{O}^\dagger}$ breaks the PRFSGs $B^{\mathcal{O}, \mathcal{O}^\dagger}$, which is the contradiction. $\square$

A.2 Black-Box Construction Relative to Oracles

In Definition 1.2, we defined a black-box construction from PRFSGs to PRUs. Similarly, we can define a black-box construction for the general cryptographic primitives as follows.

Definition A.1 (Black-Box Construction [CM24, CCS24]). *We say that a primitive $\mathcal{Q}$ can be constructed from a primitive $\mathcal{P}$ in a black-box way if there exist QPT algorithms $C^{(\cdot, \cdot)}$ and $R^{(\cdot, \cdot)}$ such that*

1. Black-box construction: *For any QPT algorithm G satisfying the correctness of $\mathcal{P}$ and any its unitary implementation $\tilde{G}$, $C^{\tilde{G}, \tilde{G}^\dagger}$ satisfies the correctness of primitive $\mathcal{Q}$.*
2. Black-box security reduction: *For any QPT algorithm G satisfying the correctness of $\mathcal{P}$, any its unitary implementation $\tilde{G}$, any adversary $\mathcal{A}$ that breaks the $\mathcal{P}$'s security of $C^{\tilde{G}, \tilde{G}^\dagger}$, and any unitary implementation $\tilde{A}$ of $\mathcal{A}$, it holds that $R^{\tilde{A}, \tilde{A}^\dagger}$ breaks the $\mathcal{Q}$'s security of G .*

The following is shown in [CM24, CCS24].

Theorem A.2 ([CM24, CCS24]). *Suppose that there exists a black-box construction from primitive $\mathcal{P}$ to $\mathcal{Q}$. Then, for any unitary $\mathcal{O}$, if $\mathcal{P}$ exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$, $\mathcal{Q}$ also exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$.*

By combing Theorem A.2 and Theorem 3.3, we have the following.

²⁵In general G is a CPTP map. The CPTP map G can be implemented by applying a unitary $\tilde{G}$ on a state and tracing out some qubits. A unitary implementation of G is such a unitary $\tilde{G}$.

Theorem A.3. *With probability 1 over the choice of $\mathcal{O}$ defined in Definition 3.1, PRSGs, IND-CPA SKE, EUF-CMA MAC with unclonable tags, UPSGs, private-key money scheme, OWSGs, OWpuzzs, and EFI exist relative to $\mathcal{O}$ and $\mathcal{O}^\dagger$.*

Proof. From the previous works, there are black-box constructions from PRFGs to them. Thus, from Theorem A.2 and Theorem 3.3, we obtain the above claim. $\square$

Therefore, from the above theorem, we have the following. We omit its proof because we can show it by the same argument in the proof of Theorem 1.1.

Theorem A.4. *There is no black-box construction of non-adaptive and $O(\log \lambda)$ -ancilla PRUs from PRSGs, IND-CPA SKE, EUF-CMA MAC with unclonable tags, UPSGs, private-key money scheme, OWSGs, OWpuzzs, or EFI.*